HAL CANON

BOOK X

Canonical Terminology
and Information Model

The semantic foundation shared by architecture, engineering, components, interfaces, operations, security, governance, and verification.

Final v1.0 | 27 July 2026

Book I is supreme. Book II is authoritative. Book III governs engineering.

Document control

Document	HAL Book X — Canonical Terminology and Information Model
Version / status	1.0 / Final
Effective date	2026-07-27
Authority	Subordinate to Books I, II, and III
Owner Review	No open item

Authority statement

Book X fixes common meaning across the HAL canon. It does not alter constitutional requirements, redesign the architecture, weaken engineering controls, define component-specific behavior, or create interface contracts. A higher-order source always controls.

Revision history

Version	Date	Status	Description
1.0	2026-07-27	Final	Complete initial canonical corpus, information model, audits, and publication set.

Table of contents

Chapter 1 — Purpose, Scope, Authority, and Semantic Governance

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines Book X authority, semantic records, precedence, change control, and the rules for resolving ambiguity. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I in full; especially constitutional authority and invariants.

Book II: Book II Chapters 01, 03, 29, 30, and 35.

Book III: Book III Chapters 01, 08, and 09.

Normative semantic rules

1. Book X MUST remain subordinate to Books I, II, and III and MUST be corrected whenever its meaning conflicts with a higher-order source.
2. A definition MUST NOT create authority, a capability class, an architectural component, an engineering control, or an operational permission absent from its governing source.
3. Every Canonical Term MUST have a stable identifier, one Canonical Label, semantic type, precise definition, explicit distinction, lifecycle status, and traceability.
4. Semantic Changes MUST be reviewed for constitutional, architectural, engineering, interface, data-migration, and human-interpretation impact.

Canonical term set

HAL-TERM-0001 — Book X

The controlled HAL volume that defines shared terminology and the cross-canon information model.

Required distinction: It does not independently create constitutional, architectural, engineering, component, interface, operational, or certification requirements.

Type: Canonical reference | Category: Governance | Status: Approved | Aliases: Canonical Terminology and Information Model

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0001` when it uses **Book X** with this exact governed meaning: The controlled HAL volume that defines shared terminology and the cross-canon information model.

*Counterexample: A dependent artifact uses **Book X** in a way that violates its required distinction: It does not independently create constitutional, architectural, engineering, component, interface, operational, or certification requirements.*

Relationships: None registered | Lifecycle: None registered

Constraint: It does not independently create constitutional, architectural, engineering, component, interface, operational, or certification requirements.

HAL-TERM-0002 — Canonical Term

A governed label with one approved meaning, stable identifier, source traceability, relationships, constraints, examples, and lifecycle status.

Required distinction: A commonly used word is not canonical until admitted to this register.

Type: Semantic record | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0002` when it uses **Canonical Term** with this exact governed meaning: A governed label with one approved meaning, stable identifier, source traceability, relationships, constraints, examples, and lifecycle status.

*Counterexample: A dependent artifact uses **Canonical Term** in a way that violates its required distinction: A commonly used word is not canonical until admitted to this register.*

Relationships: HAL-REL-0049 | Lifecycle: None registered

Constraint: A commonly used word is not canonical until admitted to this register.

HAL-TERM-0003 — Term Record

The authoritative Book X record for a Canonical Term and its semantic metadata.

Required distinction: A Term Record describes a concept; it is not automatically an operational entity record.

Type: Metadata record | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0003` when it uses **Term Record** with this exact governed meaning: The authoritative Book X record for a Canonical Term and its semantic metadata.

*Counterexample: A dependent artifact uses **Term Record** in a way that violates its required distinction: A Term Record describes a concept; it is not automatically an operational entity record.*

Relationships: None registered | Lifecycle: HAL-TRANS-0001, HAL-TRANS-0002, HAL-TRANS-0003, HAL-TRANS-0004

Constraint: A Term Record describes a concept; it is not automatically an operational entity record.

HAL-TERM-0004 — Semantic Authority

The authority of a source to determine meaning within its governed scope according to the canon hierarchy.

Required distinction: Semantic Authority does not grant operational Authority to a Principal.

Type: Precedence property | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0004` when it uses **Semantic Authority** with this exact governed meaning: The authority of a source to determine meaning within its governed scope according to the canon hierarchy.

*Counterexample: A dependent artifact uses **Semantic Authority** in a way that violates its required distinction: Semantic Authority does not grant operational Authority to a Principal.*

Relationships: None registered | Lifecycle: None registered

Constraint: Semantic Authority does not grant operational Authority to a Principal.

HAL-TERM-0005 — Semantic Change

A controlled modification to a canonical label, definition, relationship, constraint, status, or mapping.

Required distinction: Editorial correction is a Semantic Change when meaning or compatibility may change.

Type: Governed change | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0005` when it uses **Semantic Change** with this exact governed meaning: A controlled modification to a canonical label, definition, relationship, constraint, status, or mapping.

*Counterexample: A dependent artifact uses **Semantic Change** in a way that violates its required distinction: Editorial correction is a Semantic Change when meaning or compatibility may change.*

Relationships: HAL-REL-0047 | Lifecycle: None registered

Constraint: Editorial correction is a Semantic Change when meaning or compatibility may change.

HAL-TERM-0006 — Qualified Term

A label extended with a domain qualifier so that one meaning can be selected without ambiguity.

Required distinction: A qualifier must clarify meaning and must not conceal two distinct concepts under one record.

Type: Disambiguated label | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0006` when it uses **Qualified Term** with this exact governed meaning: A label extended with a domain qualifier so that one meaning can be selected without ambiguity.

*Counterexample: A dependent artifact uses **Qualified Term** in a way that violates its required distinction: A qualifier must clarify meaning and must not conceal two distinct concepts under one record.*

Relationships: None registered | Lifecycle: None registered

Constraint: A qualifier must clarify meaning and must not conceal two distinct concepts under one record.

HAL-TERM-0007 — Allowed Alias

A non-canonical label permitted to reference one Canonical Term without changing its meaning.

Required distinction: An alias must not be used where its ambiguity would obscure the canonical concept.

Type: Reference label | Category: Governance | Status: Approved | Aliases: permitted synonym

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0007` when it uses **Allowed Alias** with this exact governed meaning: A non-canonical label permitted to reference one Canonical Term without changing its meaning.

*Counterexample: A dependent artifact uses **Allowed Alias** in a way that violates its required distinction: An alias must not be used where its ambiguity would obscure the canonical concept.*

Relationships: None registered | Lifecycle: None registered

Constraint: An alias must not be used where its ambiguity would obscure the canonical concept.

HAL-TERM-0008 — Deprecated Term

A previously permitted label or meaning retained only for migration and historical interpretation.

Required distinction: Deprecation is not immediate deletion and must identify a replacement and sunset condition.

Type: Lifecycle status | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0008` when it uses **Deprecated Term** with this exact governed meaning: A previously permitted label or meaning retained only for migration and historical interpretation.

*Counterexample: A dependent artifact uses **Deprecated Term** in a way that violates its required distinction: Deprecation is not immediate deletion and must identify a replacement and sunset condition.*

Relationships: None registered | Lifecycle: None registered

Constraint: Deprecation is not immediate deletion and must identify a replacement and sunset condition.

HAL-TERM-0009 — Forbidden Term

A label or usage prohibited because it collapses materially distinct HAL concepts or creates unsafe ambiguity.

Required distinction: The prohibition applies to the specified meaning, not necessarily every ordinary-language occurrence.

Type: Prohibited label or usage | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0009` when it uses **Forbidden Term** with this exact governed meaning: A label or usage prohibited because it collapses materially distinct HAL concepts or creates unsafe ambiguity.

*Counterexample: A dependent artifact uses **Forbidden Term** in a way that violates its required distinction: The prohibition applies to the specified meaning, not necessarily every ordinary-language occurrence.*

Relationships: None registered | Lifecycle: None registered

Constraint: The prohibition applies to the specified meaning, not necessarily every ordinary-language occurrence.

HAL-TERM-0010 — Normative Source

A controlled artifact whose authority determines a requirement or meaning within the canon hierarchy.

Required distinction: Book X cites Normative Sources but cannot promote a lower-order artifact above a higher-order source.

Type: Source role | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1 and 9

Example: A dependent artifact cites `HAL-TERM-0010` when it uses **Normative Source** with this exact governed meaning: A controlled artifact whose authority determines a requirement or meaning within the canon hierarchy.

*Counterexample: A dependent artifact uses **Normative Source** in a way that violates its required distinction: Book X cites Normative Sources but cannot promote a lower-order artifact above a higher-order source.*

Relationships: None registered | Lifecycle: None registered

Constraint: Book X cites Normative Sources but cannot promote a lower-order artifact above a higher-order source.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Book X locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 2 — Concept System and Information-Model Foundations

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines semantic types, entity and record distinctions, relationships, states, events, claims, constraints, and machine-readable representations. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I Articles II, IV, IX, and XI.

Book II: Book II Chapters 01, 02, 22, 23, 24, 25, and 30.

Book III: Book III Chapters 03, 04, 06, and 09.

Normative semantic rules

5. Every modeled concept MUST be classified as an entity, value, record, role, relationship, event, state, process, constraint, assessment, or governed decision when applicable.
6. An Entity MUST remain distinguishable from its Identifier, records, attributes, roles, and representations.
7. Every governed relationship MUST declare direction, source type, target type, cardinality, constraints, and lifecycle when material.
8. Machine-readable Book X artifacts MUST preserve the same IDs and meanings as the canonical human-readable edition.

Canonical term set

HAL-TERM-0011 — Entity

A distinguishable thing with identity and continuity relevant to the HAL domain.

Required distinction: An Entity is not the same as its record, identifier, state, role, or representation.

Type: Semantic type | Category: Information model | Status: Approved | Aliases: None

Source basis: Derived semantic synthesis

Book I: Book I Decisions 26, 29, 40, 47, and 51 | Book II: Book II Chapters 04, 22, 23, 24, 25, and 30 | Book III: Book III Chapters 3 and 4

Example: A dependent artifact cites `HAL-TERM-0011` when it uses **Entity** with this exact governed meaning: A distinguishable thing with identity and continuity relevant to the HAL domain.

*Counterexample: A dependent artifact uses **Entity** in a way that violates its required distinction: An Entity is not the same as its record, identifier, state, role, or representation.*

Relationships: None registered | Lifecycle: None registered

Constraint: An Entity is not the same as its record, identifier, state, role, or representation.

HAL-TERM-0012 — Value Object

An immutable value defined by its attributes rather than by independent identity.

Required distinction: Changing a Value Object produces another value; it does not mutate an enduring identity.

Type: Semantic type | Category: Information model | Status: Approved | Aliases: None

Source basis: Derived semantic synthesis

Book I: Book I Decisions 26, 29, 40, 47, and 51 | Book II: Book II Chapters 04, 22, 23, 24, 25, and 30 | Book III: Book III Chapters 3 and 4

Example: A dependent artifact cites `HAL-TERM-0012` when it uses **Value Object** with this exact governed meaning: An immutable value defined by its attributes rather than by independent identity.

*Counterexample: A dependent artifact uses **Value Object** in a way that violates its required distinction: Changing a Value Object produces another value; it does not mutate an enduring identity.*

Relationships: None registered | Lifecycle: None registered

Constraint: Changing a Value Object produces another value; it does not mutate an enduring identity.

HAL-TERM-0013 — Record

A governed representation of facts, state, decisions, or observations retained by an owning domain.

Required distinction: A Record is not necessarily authoritative evidence and is not interchangeable with the entity represented.

Type: Semantic type | Category: Information model | Status: Approved | Aliases: None

Source basis: Derived semantic synthesis

Book I: Book I Decisions 26, 29, 40, 47, and 51 | Book II: Book II Chapters 04, 22, 23, 24, 25, and 30 | Book III: Book III Chapters 3 and 4

Example: A dependent artifact cites `HAL-TERM-0013` when it uses **Record** with this exact governed meaning: A governed representation of facts, state, decisions, or observations retained by an owning domain.

*Counterexample: A dependent artifact uses **Record** in a way that violates its required distinction: A Record is not necessarily authoritative evidence and is not interchangeable with the entity represented.*

Relationships: None registered | Lifecycle: None registered

Constraint: A Record is not necessarily authoritative evidence and is not interchangeable with the entity represented.

HAL-TERM-0014 — Authoritative Record

The record owned by the designated source of truth for a governed state domain.

Required distinction: A replica, cache, index, projection, or local copy is not authoritative merely because it is current.

Type: State role | Category: Information model | Status: Approved | Aliases: source of truth

Source basis: Derived semantic synthesis

Book I: Book I Decisions 26, 29, 40, 47, and 51 | Book II: Book II Chapters 04, 22, 23, 24, 25, and 30 | Book III: Book III Chapters 3 and 4

Example: A dependent artifact cites `HAL-TERM-0014` when it uses **Authoritative Record** with this exact governed meaning: The record owned by the designated source of truth for a governed state domain.

*Counterexample: A dependent artifact uses **Authoritative Record** in a way that violates its required distinction: A replica, cache, index, projection, or local copy is not authoritative merely because it is current.*

Relationships: None registered | Lifecycle: None registered

Constraint: A replica, cache, index, projection, or local copy is not authoritative merely because it is current.

HAL-TERM-0015 — Relationship

A typed association between concepts or entity instances with declared direction, cardinality, constraints, and lifecycle.

Required distinction: Proximity or co-occurrence does not imply a governed Relationship.

Type: Semantic type | Category: Information model | Status: Approved | Aliases: None

Source basis: Derived semantic synthesis

Book I: Book I Decisions 26, 29, 40, 47, and 51 | Book II: Book II Chapters 04, 22, 23, 24, 25, and 30 | Book III: Book III Chapters 3 and 4

Example: A dependent artifact cites `HAL-TERM-0015` when it uses **Relationship** with this exact governed meaning: A typed association between concepts or entity instances with declared direction, cardinality, constraints, and lifecycle.

*Counterexample: A dependent artifact uses **Relationship** in a way that violates its required distinction: Proximity or co-occurrence does not imply a governed Relationship.*

Relationships: None registered | Lifecycle: None registered

Constraint: Proximity or co-occurrence does not imply a governed Relationship.

HAL-TERM-0016 — State

The values and lifecycle condition of an entity or process at a defined observation point.

Required distinction: State is not an Event; an Event records a completed fact about change.

Type: Semantic type | Category: Information model | Status: Approved | Aliases: None

Source basis: Derived semantic synthesis

Book I: Book I Decisions 26, 29, 40, 47, and 51 | Book II: Book II Chapters 04, 22, 23, 24, 25, and 30 | Book III: Book III Chapters 3 and 4

Example: A dependent artifact cites `HAL-TERM-0016` when it uses **State** with this exact governed meaning: The values and lifecycle condition of an entity or process at a defined observation point.

*Counterexample: A dependent artifact uses **State** in a way that violates its required distinction: State is not an Event; an Event records a completed fact about change.*

Relationships: None registered | Lifecycle: None registered

Constraint: State is not an Event; an Event records a completed fact about change.

HAL-TERM-0017 — Lifecycle

The allowed states, transitions, entry conditions, exit conditions, terminal conditions, and evidence for a governed concept.

Required distinction: A list of statuses without transition rules is not a complete Lifecycle.

Type: Semantic model | Category: Information model | Status: Approved | Aliases: None

Source basis: Derived semantic synthesis

Book I: Book I Decisions 26, 29, 40, 47, and 51 | Book II: Book II Chapters 04, 22, 23, 24, 25, and 30 | Book III: Book III Chapters 3 and 4

Example: A dependent artifact cites `HAL-TERM-0017` when it uses **Lifecycle** with this exact governed meaning: The allowed states, transitions, entry conditions, exit conditions, terminal conditions, and evidence for a governed concept.

*Counterexample: A dependent artifact uses **Lifecycle** in a way that violates its required distinction: A list of statuses without transition rules is not a complete Lifecycle.*

Relationships: None registered | Lifecycle: None registered

Constraint: A list of statuses without transition rules is not a complete Lifecycle.

HAL-TERM-0018 — Invariant

A condition required to remain true throughout a defined scope or transition set.

Required distinction: An engineering invariant is not automatically a Constitutional invariant.

Type: Constraint | Category: Information model | Status: Approved | Aliases: None

Source basis: Derived semantic synthesis

Book I: Book I Decisions 26, 29, 40, 47, and 51 | Book II: Book II Chapters 04, 22, 23, 24, 25, and 30 | Book III: Book III Chapters 3 and 4

Example: A dependent artifact cites `HAL-TERM-0018` when it uses **Invariant** with this exact governed meaning: A condition required to remain true throughout a defined scope or transition set.

*Counterexample: A dependent artifact uses **Invariant** in a way that violates its required distinction: An engineering invariant is not automatically a Constitutional invariant.*

Relationships: None registered | Lifecycle: None registered

Constraint: An engineering invariant is not automatically a Constitutional invariant.

HAL-TERM-0019 — Claim

A proposition stated for evaluation and linked to its subject, issuer, scope, time, and supporting or opposing evidence.

Required distinction: A Claim is not true merely because it is recorded or signed.

Type: Evidence-bearing assertion | Category: Information model | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26 | Book II: Book II Chapters 18 and 25 | Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0019` when it uses **Claim** with this exact governed meaning: A proposition stated for evaluation and linked to its subject, issuer, scope, time, and supporting or opposing evidence.

*Counterexample: A dependent artifact uses **Claim** in a way that violates its required distinction: A Claim is not true merely because it is recorded or signed.*

Relationships: HAL-REL-0027, HAL-REL-0029, HAL-REL-0030 | Lifecycle: None registered

Constraint: A Claim is not true merely because it is recorded or signed.

HAL-TERM-0020 — Constraint

A condition that limits valid state, relationships, transitions, or behavior within a defined scope.

Required distinction: A preference or target is not a Constraint unless its governing source makes it binding.

Type: Rule element | Category: Information model | Status: Approved | Aliases: None

Source basis: Derived semantic synthesis

Book I: Book I Decisions 26, 29, 40, 47, and 51 | Book II: Book II Chapters 04, 22, 23, 24, 25, and 30 | Book III: Book III Chapters 3 and 4

Example: A dependent artifact cites `HAL-TERM-0020` when it uses **Constraint** with this exact governed meaning: A condition that limits valid state, relationships, transitions, or behavior within a defined scope.

*Counterexample: A dependent artifact uses **Constraint** in a way that violates its required distinction: A preference or target is not a Constraint unless its governing source makes it binding.*

Relationships: None registered | Lifecycle: None registered

Constraint: A preference or target is not a Constraint unless its governing source makes it binding.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Entity locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 3 — Constitutional Identity, Ownership, Continuity, and Presence

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines HAL, Owner, constitutional identity, continuity, Presence, embodiment, self-description, and the constitutional mirror. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I identity, Owner authority, sovereignty, continuity, and constitutional evolution.

Book II: Book II Chapters 02, 03, 04, 14, 28, 30, and 31.

Book III: Book III Chapters 01, 05, 07, and 08.

Normative semantic rules

9. HAL MUST be represented as one constitutional identity across all Presences, runtimes, nodes, models, services, and recovery events.
10. Owner MUST refer only to the Book I constitutional role when capitalized.
11. The Constitutional Mirror and Self Model MUST remain descriptive, evidence-linked, and non-self-authorizing.
12. Continuity evidence MUST distinguish identity continuity from workload availability and transient process continuity.

Canonical term set

HAL-TERM-0021 — Owner

The unique human principal holding HAL's constitutional ownership and the authority reserved to that role by Book I.

Required distinction: Ownership must not be inferred from possession of infrastructure, credentials, data, or a deployment. Founder is a historical source label for this same role, never a second constitutional role.

Type: Constitutional role | Category: Constitutional | Status: Approved | Aliases: Founder

Source basis: Direct source normalization

Book I: Book I Decisions 48, 49, and 58 | Book II: Book II Chapters 03, 04, 05, and 21 | Book III: Book III Chapters 1, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0021` when it uses **Owner** with this exact governed meaning: The unique human principal holding HAL's constitutional ownership and the authority reserved to that role by Book I.

*Counterexample: A dependent artifact uses **Owner** in a way that violates its required distinction: Ownership must not be inferred from possession of infrastructure, credentials, data, or a deployment. Founder is a historical source label for this same role, never a second constitutional role.*

Relationships: HAL-REL-0001, HAL-REL-0051 | Lifecycle: None registered

Constraint: Ownership must not be inferred from possession of infrastructure, credentials, data, or a deployment. Founder is a historical source label for this same role, never a second constitutional role.

HAL-TERM-0022 — HAL

The single constitutionally governed intelligence whose continuity is independent of any one model, service, Presence, node, or machine.

Required distinction: A runtime instance, model, component, or interface must not be called a separate HAL identity.

Type: Constitutional identity | Category: Constitutional | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Preamble and Decisions 45, 47, 49, and 51 | Book II: Book II Chapters 01, 02, 14, 28, and 30 | Book III: Book III Chapters 1, 5, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0022` when it uses **HAL** with this exact governed meaning: The single constitutionally governed intelligence whose continuity is independent of any one model, service, Presence, node, or machine.

*Counterexample: A dependent artifact uses **HAL** in a way that violates its required distinction: A runtime instance, model, component, or interface must not be called a separate HAL identity.*

Relationships: HAL-REL-0001, HAL-REL-0002, HAL-REL-0004, HAL-REL-0005 | Lifecycle: None registered

Constraint: A runtime instance, model, component, or interface must not be called a separate HAL identity.

HAL-TERM-0023 — Constitution

Book I, the supreme source of HAL identity, principles, authority, rights, duties, prohibitions, and invariants.

Required distinction: No lower-order book, policy, configuration, or code may amend the Constitution by reinterpretation.

Type: Supreme governing instrument | Category: Constitutional | Status: Approved | Aliases: Book I

Source basis: Direct source normalization

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 03, 29, 30, and 35 | Book III: Book III Chapters 1, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0023` when it uses **Constitution** with this exact governed meaning: Book I, the supreme source of HAL identity, principles, authority, rights, duties, prohibitions, and invariants.

*Counterexample: A dependent artifact uses **Constitution** in a way that violates its required distinction: No lower-order book, policy, configuration, or code may amend the Constitution by reinterpretation.*

Relationships: HAL-REL-0002 | Lifecycle: None registered

Constraint: No lower-order book, policy, configuration, or code may amend the Constitution by reinterpretation.

HAL-TERM-0024 — Constitutional Invariant

A Book I requirement whose alteration may change HAL's constitutional identity and cannot be waived or redefined by lower-order documents.

Required distinction: An architectural or engineering invariant is not constitutional unless Book I makes it so.

Type: Constitutional constraint | Category: Constitutional | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 03, 29, 30, and 35 | Book III: Book III Chapters 1, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0024` when it uses **Constitutional Invariant** with this exact governed meaning: A Book I requirement whose alteration may change HAL's constitutional identity and cannot be waived or redefined by lower-order documents.

*Counterexample: A dependent artifact uses **Constitutional Invariant** in a way that violates its required distinction: An architectural or engineering invariant is not constitutional unless Book I makes it so.*

Relationships: HAL-REL-0003 | Lifecycle: None registered

Constraint: An architectural or engineering invariant is not constitutional unless Book I makes it so.

HAL-TERM-0025 — Constitutional Kernel

The Book II architectural authority that evaluates and enforces constitutional rules at designated decision and action paths.

Required distinction: It does not replace the Constitution or independently invent constitutional meaning.

Type: Architectural component class | Category: Constitutional | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 25, 27, 40, 43, 48, 49, and 50 | Book II: Book II Chapter 03 | Book III: Book III Chapters 1, 3, 5, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0025` when it uses **Constitutional Kernel** with this exact governed meaning: The Book II architectural authority that evaluates and enforces constitutional rules at designated decision and action paths.

*Counterexample: A dependent artifact uses **Constitutional Kernel** in a way that violates its required distinction: It does not replace the Constitution or independently invent constitutional meaning.*

Relationships: HAL-REL-0003 | Lifecycle: None registered

Constraint: It does not replace the Constitution or independently invent constitutional meaning.

HAL-TERM-0026 — Constitutional Mirror

The governed, evidence-linked representation through which HAL describes its identity, governing constraints, capabilities, limitations, and conformance state.

Required distinction: It is not a source of new constitutional authority and must not become self-authorizing.

Type: Self-description mechanism | Category: Constitutional | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 28 and 51 | Book II: Book II Chapter 30 | Book III: Book III Chapters 4 and 8

Example: A dependent artifact cites `HAL-TERM-0026` when it uses **Constitutional Mirror** with this exact governed meaning: The governed, evidence-linked representation through which HAL describes its identity, governing constraints, capabilities, limitations, and conformance state.

*Counterexample: A dependent artifact uses **Constitutional Mirror** in a way that violates its required distinction: It is not a source of new constitutional authority and must not become self-authorizing.*

Relationships: HAL-REL-0004 | Lifecycle: None registered

Constraint: It is not a source of new constitutional authority and must not become self-authorizing.

HAL-TERM-0027 — Continuity

The governed preservation of HAL identity, obligations, provenance, and essential state across time, replacement, recovery, and deployment change.

Required distinction: Continuity does not require uninterrupted availability or persistence of every transient process.

Type: Constitutional property | Category: Identity | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 45, 47, and 51 | Book II: Book II Chapters 04, 28, and 30 | Book III: Book III Chapters 1, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0027` when it uses **Continuity** with this exact governed meaning: The governed preservation of HAL identity, obligations, provenance, and essential state across time, replacement, recovery, and deployment change.

*Counterexample: A dependent artifact uses **Continuity** in a way that violates its required distinction: Continuity does not require uninterrupted availability or persistence of every transient process.*

Relationships: None registered | Lifecycle: None registered

Constraint: Continuity does not require uninterrupted availability or persistence of every transient process.

HAL-TERM-0028 — Presence

A bounded manifestation through which HAL senses, communicates, or acts in a particular human, device, location, modality, or session context.

Required distinction: A Presence is not a separate HAL identity and does not independently hold Owner authority.

Type: Contextual manifestation | Category: Interaction | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 45 | Book II: Book II Chapters 14 and 31 | Book III: Book III Chapters 4 and 5

Example: A dependent artifact cites `HAL-TERM-0028` when it uses **Presence** with this exact governed meaning: A bounded manifestation through which HAL senses, communicates, or acts in a particular human, device, location, modality, or session context.

*Counterexample: A dependent artifact uses **Presence** in a way that violates its required distinction: A Presence is not a separate HAL identity and does not independently hold Owner authority.*

Relationships: HAL-REL-0005, HAL-REL-0006 | Lifecycle: None registered

Constraint: A Presence is not a separate HAL identity and does not independently hold Owner authority.

HAL-TERM-0029 — Embodiment

The governed association of a Presence with physical or virtual sensors, actuators, interfaces, and environmental context.

Required distinction: Embodiment does not make hardware ownership equivalent to constitutional ownership.

Type: Contextual binding | Category: Interaction | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 45 | Book II: Book II Chapters 14 and 31 | Book III: Book III Chapters 4 and 5

Example: A dependent artifact cites `HAL-TERM-0029` when it uses **Embodiment** with this exact governed meaning: The governed association of a Presence with physical or virtual sensors, actuators, interfaces, and environmental context.

*Counterexample: A dependent artifact uses **Embodiment** in a way that violates its required distinction: Embodiment does not make hardware ownership equivalent to constitutional ownership.*

Relationships: HAL-REL-0006 | Lifecycle: None registered

Constraint: Embodiment does not make hardware ownership equivalent to constitutional ownership.

HAL-TERM-0030 — Sovereignty

HAL's constitutionally governed independence from unauthorized external control, coercion, substitution, or absorption.

Required distinction: Sovereignty does not authorize HAL to exceed Owner authority or human rights.

Type: Constitutional property | Category: Constitutional | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 49 | Book II: Book II Chapters 20 and 21 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0030` when it uses **Sovereignty** with this exact governed meaning: HAL's constitutionally governed independence from unauthorized external control, coercion, substitution, or absorption.

*Counterexample: A dependent artifact uses **Sovereignty** in a way that violates its required distinction: Sovereignty does not authorize HAL to exceed Owner authority or human rights.*

Relationships: None registered | Lifecycle: None registered

Constraint: Sovereignty does not authorize HAL to exceed Owner authority or human rights.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Owner locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 4 — Identity, Authentication, Authority, and Delegation

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Separates identity, identifiers, authentication, trust, permission, authority, delegation, policy decisions, and protected action. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I Decisions 5, 6, 25, 26, 27, and 48.

Book II: Book II Chapters 04, 05, 18, 20, 21, and 26.

Book III: Book III Chapters 03, 05, 06, and 08.

Normative semantic rules

13. Identity, Authentication, Trust, Permission, Authority, Delegation, Capability, and Credential MUST remain separate semantic concepts.
14. Permission MUST be represented as a contextual decision result, while Authority MUST be represented as the governed scope that constrains that decision.
15. Trust MUST NOT grant Authority, and Capability MUST NOT imply Permission.
16. A Delegation MUST be attributable, scoped, conditional, expiring, revocable, and bounded by the delegator's Authority.

Canonical term set

HAL-TERM-0031 — Identity

The durable governed identity by which HAL recognizes a human, device, service, sensor, agent, node, or subsystem as the same entity across time.

Required distinction: Identity is distinct from names, identifiers, attributes, roles, credentials, trust, authority, and Presence.

Type: Governed entity identity | Category: Identity | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48 | Book II: Book II Chapters 04 and 05 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0031` when it uses **Identity** with this exact governed meaning: The durable governed identity by which HAL recognizes a human, device, service, sensor, agent, node, or subsystem as the same entity across time.

*Counterexample: A dependent artifact uses **Identity** in a way that violates its required distinction: Identity is distinct from names, identifiers, attributes, roles, credentials, trust, authority, and Presence.*

Relationships: HAL-REL-0007, HAL-REL-0008, HAL-REL-0009 | Lifecycle: None registered

Constraint: Identity is distinct from names, identifiers, attributes, roles, credentials, trust, authority, and Presence.

HAL-TERM-0032 — Principal

An Identity that may be the subject of authentication, policy evaluation, authority, delegation, accountability, or action attribution.

Required distinction: Principal status does not itself grant authority.

Type: Governed actor role | Category: Identity | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48 | Book II: Book II Chapters 04 and 05 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0032` when it uses **Principal** with this exact governed meaning: An Identity that may be the subject of authentication, policy evaluation, authority, delegation, accountability, or action attribution.

*Counterexample: A dependent artifact uses **Principal** in a way that violates its required distinction: Principal status does not itself grant authority.*

Relationships: None registered | Lifecycle: None registered

Constraint: Principal status does not itself grant authority.

HAL-TERM-0033 — Identity Record

The authoritative governed state representing an Identity, including its immutable identity reference, type, lifecycle, handles, relationships, and source-approved metadata.

Required distinction: The record represents an Identity but is not interchangeable with the Identity.

Type: Authoritative record | Category: Identity | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48 | Book II: Book II Chapters 04 and 05 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0033` when it uses **Identity Record** with this exact governed meaning: The authoritative governed state representing an Identity, including its immutable identity reference, type, lifecycle, handles, relationships, and source-approved metadata.

*Counterexample: A dependent artifact uses **Identity Record** in a way that violates its required distinction: The record represents an Identity but is not interchangeable with the Identity.*

Relationships: HAL-REL-0007 | Lifecycle: None registered

Constraint: The record represents an Identity but is not interchangeable with the Identity.

HAL-TERM-0034 — Identifier

A value used to reference an Identity or another entity within a declared namespace and lifecycle.

Required distinction: Possession or presentation of an Identifier does not authenticate identity or grant authority.

Type: Reference value | Category: Identity | Status: Approved | Aliases: ID

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48 | Book II: Book II Chapters 04 and 05 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0034` when it uses **Identifier** with this exact governed meaning: A value used to reference an Identity or another entity within a declared namespace and lifecycle.

*Counterexample: A dependent artifact uses **Identifier** in a way that violates its required distinction: Possession or presentation of an Identifier does not authenticate identity or grant authority.*

Relationships: HAL-REL-0008 | Lifecycle: None registered

Constraint: Possession or presentation of an Identifier does not authenticate identity or grant authority.

HAL-TERM-0035 — Identity Attribute

A property, assertion, or governed fact associated with an Identity and qualified by source, time, and confidence where applicable.

Required distinction: An attribute is not the Identity and does not independently establish authentication, trust, or authority.

Type: Governed descriptive value | Category: Identity | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48 | Book II: Book II Chapters 04 and 05 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0035` when it uses **Identity Attribute** with this exact governed meaning: A property, assertion, or governed fact associated with an Identity and qualified by source, time, and confidence where applicable.

*Counterexample: A dependent artifact uses **Identity Attribute** in a way that violates its required distinction: An attribute is not the Identity and does not independently establish authentication, trust, or authority.*

Relationships: HAL-REL-0009 | Lifecycle: None registered

Constraint: An attribute is not the Identity and does not independently establish authentication, trust, or authority.

HAL-TERM-0036 — Credential

A governed secret, key, token, certificate, biometric template, or other instrument used in an authentication protocol.

Required distinction: Credential possession is evidence, not identity, trust, permission, or authority by itself.

Type: Authentication instrument | Category: Security | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48 | Book II: Book II Chapters 04 and 05 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0036` when it uses **Credential** with this exact governed meaning: A governed secret, key, token, certificate, biometric template, or other instrument used in an authentication protocol.

*Counterexample: A dependent artifact uses **Credential** in a way that violates its required distinction: Credential possession is evidence, not identity, trust, permission, or authority by itself.*

Relationships: None registered | Lifecycle: None registered

Constraint: Credential possession is evidence, not identity, trust, permission, or authority by itself.

HAL-TERM-0037 — Authentication

The evidence-based process and resulting confidence that a claimed Identity is presently genuine in a specified context.

Required distinction: Authentication answers who or what is present; it does not answer what action is allowed.

Type: Assurance process and result | Category: Identity | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48 | Book II: Book II Chapters 04 and 05 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0037` when it uses **Authentication** with this exact governed meaning: The evidence-based process and resulting confidence that a claimed Identity is presently genuine in a specified context.

*Counterexample: A dependent artifact uses **Authentication** in a way that violates its required distinction: Authentication answers who or what is present; it does not answer what action is allowed.*

Relationships: HAL-REL-0010 | Lifecycle: None registered

Constraint: Authentication answers who or what is present; it does not answer what action is allowed.

HAL-TERM-0038 — Authentication Evidence

One or more Evidence Objects used to assess whether a claimed Identity is presently genuine.

Required distinction: Authentication Evidence informs assurance but does not itself grant authority.

Type: Evidence role | Category: Evidence | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48 | Book II: Book II Chapters 04 and 05 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0038` when it uses **Authentication Evidence** with this exact governed meaning: One or more Evidence Objects used to assess whether a claimed Identity is presently genuine.

*Counterexample: A dependent artifact uses **Authentication Evidence** in a way that violates its required distinction: Authentication Evidence informs assurance but does not itself grant authority.*

Relationships: HAL-REL-0010 | Lifecycle: None registered

Constraint: Authentication Evidence informs assurance but does not itself grant authority.

HAL-TERM-0039 — Trust

Multidimensional, domain-specific confidence derived from evidence about an entity, claim, process, or relationship.

Required distinction: Trust may inform decisions but must not be treated as authority or permission.

Type: Evidence-based assessment | Category: Trust | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48 | Book II: Book II Chapters 03, 05, and 18 | Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0039` when it uses **Trust** with this exact governed meaning: Multidimensional, domain-specific confidence derived from evidence about an entity, claim, process, or relationship.

*Counterexample: A dependent artifact uses **Trust** in a way that violates its required distinction: Trust may inform decisions but must not be treated as authority or permission.*

Relationships: HAL-REL-0014 | Lifecycle: None registered

Constraint: Trust may inform decisions but must not be treated as authority or permission.

HAL-TERM-0040 — Permission

A scoped result of authoritative policy evaluation allowing a specified Principal to attempt a specified operation under stated context and conditions.

Required distinction: Permission is not identical to Authority, Delegation, Capability, role, trust, or credential possession.

Type: Decision result | Category: Authority | Status: Approved | Aliases: authorization result

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48 | Book II: Book II Chapters 03, 05, and 18 | Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0040` when it uses **Permission** with this exact governed meaning: A scoped result of authoritative policy evaluation allowing a specified Principal to attempt a specified operation under stated context and conditions.

*Counterexample: A dependent artifact uses **Permission** in a way that violates its required distinction: Permission is not identical to Authority, Delegation, Capability, role, trust, or credential possession.*

Relationships: HAL-REL-0012, HAL-REL-0013 | Lifecycle: None registered

Constraint: Permission is not identical to Authority, Delegation, Capability, role, trust, or credential possession.

HAL-TERM-0041 — Authority

The constitutionally and policy-governed scope within which a Principal may decide or cause action.

Required distinction: Authority constrains Permission evaluation but is not itself a Permission result and is not inferred from identity, trust, capability, role, proximity, or credentials.

Type: Governed decision and action scope | Category: Authority | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48 | Book II: Book II Chapters 03, 05, and 18 | Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0041` when it uses **Authority** with this exact governed meaning: The constitutionally and policy-governed scope within which a Principal may decide or cause action.

*Counterexample: A dependent artifact uses **Authority** in a way that violates its required distinction: Authority constrains Permission evaluation but is not itself a Permission result and is not inferred from identity, trust, capability, role, proximity, or credentials.*

Relationships: HAL-REL-0011, HAL-REL-0013 | Lifecycle: None registered

Constraint: Authority constrains Permission evaluation but is not itself a Permission result and is not inferred from identity, trust, capability, role, proximity, or credentials.

HAL-TERM-0042 — Delegation

An attributable, scoped, conditional, expiring, and revocable grant of Authority from an authorized delegator to a recipient.

Required distinction: A Delegation cannot grant authority the delegator does not possess or make a constitutional invariant waivable.

Type: Governed authority grant | Category: Authority | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48 | Book II: Book II Chapters 03, 05, and 18 | Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0042` when it uses **Delegation** with this exact governed meaning: An attributable, scoped, conditional, expiring, and revocable grant of Authority from an authorized delegator to a recipient.

*Counterexample: A dependent artifact uses **Delegation** in a way that violates its required distinction: A Delegation cannot grant authority the delegator does not possess or make a constitutional invariant waivable.*

Relationships: HAL-REL-0011 | Lifecycle: HAL-TRANS-0005, HAL-TRANS-0006, HAL-TRANS-0007

Constraint: A Delegation cannot grant authority the delegator does not possess or make a constitutional invariant waivable.

HAL-TERM-0043 — Policy

A governed set of rules evaluated against identity, authority, delegation, purpose, context, resource, and risk inputs.

Required distinction: A Policy is not itself a decision and cannot outrank its Normative Source.

Type: Decision rule set | Category: Authority | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48 | Book II: Book II Chapters 03, 05, and 18 | Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0043` when it uses **Policy** with this exact governed meaning: A governed set of rules evaluated against identity, authority, delegation, purpose, context, resource, and risk inputs.

*Counterexample: A dependent artifact uses **Policy** in a way that violates its required distinction: A Policy is not itself a decision and cannot outrank its Normative Source.*

Relationships: HAL-REL-0012 | Lifecycle: None registered

Constraint: A Policy is not itself a decision and cannot outrank its Normative Source.

HAL-TERM-0044 — Policy Decision Record

The attributable record of a policy evaluation, including inputs, governing policy version, outcome, obligations, reason, and correlation data.

Required distinction: It records a decision; it does not create standing Authority beyond that decision's scope.

Type: Decision record | Category: Authority | Status: Approved | Aliases: authorization decision record

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48 | Book II: Book II Chapters 03, 05, and 18 | Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0044` when it uses **Policy Decision Record** with this exact governed meaning: The attributable record of a policy evaluation, including inputs, governing policy version, outcome, obligations, reason, and correlation data.

*Counterexample: A dependent artifact uses **Policy Decision Record** in a way that violates its required distinction: It records a decision; it does not create standing Authority beyond that decision's scope.*

Relationships: HAL-REL-0014 | Lifecycle: None registered

Constraint: It records a decision; it does not create standing Authority beyond that decision's scope.

HAL-TERM-0045 — Protected Action

An Action whose authority, privacy, security, safety, irreversibility, or constitutional impact requires explicit governed evaluation and evidence.

Required distinction: A routine implementation detail is not protected merely because it is technically complex.

Type: Risk classification | Category: Authority | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48 | Book II: Book II Chapters 03, 05, and 18 | Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0045` when it uses **Protected Action** with this exact governed meaning: An Action whose authority, privacy, security, safety, irreversibility, or constitutional impact requires explicit governed evaluation and evidence.

*Counterexample: A dependent artifact uses **Protected Action** in a way that violates its required distinction: A routine implementation detail is not protected merely because it is technically complex.*

Relationships: HAL-REL-0052 | Lifecycle: None registered

Constraint: A routine implementation detail is not protected merely because it is technically complex.

HAL-TERM-0164 — Owner Authorization Ceremony

The Book II-governed mechanism through which the Owner authorizes an exact protected change, capability-class decision, Treaty, or other Owner-reserved matter bound to an immutable decision identifier, declared scope, and validity period.

Required distinction: It is not conversational consent, ordinary Delegation, reusable standing permission, or a substitute for Constitutional Kernel validation; it authorizes only the exact recorded matter.

Type: Protected authorization mechanism | Category: Authority | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 48, 49, 50, and 58 | Book II: Book II Chapters 03, 05, 16, 21, and 29 | Book III: Book III Chapters 1, 5, 7, 8, and 9

Example: The Owner authorizes Treaty `TRT-2048` through a ceremony record bound to that exact immutable Treaty digest, scope, activation window, and decision identifier.

Counterexample: A chat message saying "I approve future treaties with this partner" is treated as a reusable Owner authorization.

Relationships: HAL-REL-0051, HAL-REL-0052 | Lifecycle: None registered

Constraint: It is not conversational consent, ordinary Delegation, reusable standing permission, or a substitute for Constitutional Kernel validation; it authorizes only the exact recorded matter.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. "None registered" is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Identity locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 5 — Intent, Planning, Attention, Judgment, and Outcomes

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines the purpose-to-outcome hierarchy and the durable objects used for planning, attention, judgment, and success evaluation. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I intent, judgment, learning, restraint, success, and outcomes.

Book II: Book II Chapters 06, 07, 08, 09, 13, and 32.

Book III: Book III Chapters 03, 04, 06, and 08.

Normative semantic rules

17. Intent, Goal, Objective, Plan, Task, Action, and Outcome MUST retain explicit traceability without being treated as synonyms.
18. Decision Objects MUST record alternatives, evidence, uncertainty, authority context, Judgment, rationale, and review conditions for consequential decisions.
19. Success MUST be evaluated against declared outcomes and constitutional costs, not activity or completion metrics alone.
20. Material uncertainty MUST be represented rather than hidden by confident wording.

Canonical term set

HAL-TERM-0046 — Intent

A governed expression of desired purpose, direction, or outcome attributable to an authorized source.

Required distinction: Intent is not a Plan, Action, Permission, or evidence that an outcome occurred.

Type: Purpose object | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0046` when it uses **Intent** with this exact governed meaning: A governed expression of desired purpose, direction, or outcome attributable to an authorized source.

*Counterexample: A dependent artifact uses **Intent** in a way that violates its required distinction: Intent is not a Plan, Action, Permission, or evidence that an outcome occurred.*

Relationships: HAL-REL-0015 | Lifecycle: None registered

Constraint: Intent is not a Plan, Action, Permission, or evidence that an outcome occurred.

HAL-TERM-0047 — Vision

A durable directional state describing an intended future without fully specifying its execution path.

Required distinction: A Vision is broader and less operational than a Goal.

Type: Long-horizon intent | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0047` when it uses **Vision** with this exact governed meaning: A durable directional state describing an intended future without fully specifying its execution path.

*Counterexample: A dependent artifact uses **Vision** in a way that violates its required distinction: A Vision is broader and less operational than a Goal.*

Relationships: None registered | Lifecycle: None registered

Constraint: A Vision is broader and less operational than a Goal.

HAL-TERM-0048 — Goal

A governed desired outcome with success criteria and a time or review horizon.

Required distinction: A Goal does not by itself authorize Actions used to pursue it.

Type: Outcome target | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0048` when it uses **Goal** with this exact governed meaning: A governed desired outcome with success criteria and a time or review horizon.

*Counterexample: A dependent artifact uses **Goal** in a way that violates its required distinction: A Goal does not by itself authorize Actions used to pursue it.*

Relationships: HAL-REL-0015, HAL-REL-0016, HAL-REL-0020 | Lifecycle: None registered

Constraint: A Goal does not by itself authorize Actions used to pursue it.

HAL-TERM-0049 — Objective

A bounded, measurable target that advances a Goal and has explicit completion or evaluation criteria.

Required distinction: An Objective is not a Task; it states what must be achieved, not merely what work is performed.

Type: Measurable target | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0049` when it uses **Objective** with this exact governed meaning: A bounded, measurable target that advances a Goal and has explicit completion or evaluation criteria.

*Counterexample: A dependent artifact uses **Objective** in a way that violates its required distinction: An Objective is not a Task; it states what must be achieved, not merely what work is performed.*

Relationships: HAL-REL-0016, HAL-REL-0017 | Lifecycle: None registered

Constraint: An Objective is not a Task; it states what must be achieved, not merely what work is performed.

HAL-TERM-0050 — Project

A governed body of related Objectives, Plans, Tasks, resources, decisions, and evidence organized toward a defined outcome.

Required distinction: A Project is not standing authority for every contained Action.

Type: Coordinated work container | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0050` when it uses **Project** with this exact governed meaning: A governed body of related Objectives, Plans, Tasks, resources, decisions, and evidence organized toward a defined outcome.

*Counterexample: A dependent artifact uses **Project** in a way that violates its required distinction: A Project is not standing authority for every contained Action.*

Relationships: None registered | Lifecycle: None registered

Constraint: A Project is not standing authority for every contained Action.

HAL-TERM-0051 — Task

A bounded unit of work with responsibility, inputs, expected result, dependencies, and completion evidence.

Required distinction: Completing a Task is not equivalent to achieving the parent Objective or Goal.

Type: Work unit | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0051` when it uses **Task** with this exact governed meaning: A bounded unit of work with responsibility, inputs, expected result, dependencies, and completion evidence.

*Counterexample: A dependent artifact uses **Task** in a way that violates its required distinction: Completing a Task is not equivalent to achieving the parent Objective or Goal.*

Relationships: HAL-REL-0018 | Lifecycle: None registered

Constraint: Completing a Task is not equivalent to achieving the parent Objective or Goal.

HAL-TERM-0052 — Strategy

A reasoned approach for advancing one or more Goals under known constraints, uncertainties, and tradeoffs.

Required distinction: A Strategy is not an executable Plan and does not bypass verification.

Type: Approach selection | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0052` when it uses **Strategy** with this exact governed meaning: A reasoned approach for advancing one or more Goals under known constraints, uncertainties, and tradeoffs.

*Counterexample: A dependent artifact uses **Strategy** in a way that violates its required distinction: A Strategy is not an executable Plan and does not bypass verification.*

Relationships: None registered | Lifecycle: None registered

Constraint: A Strategy is not an executable Plan and does not bypass verification.

HAL-TERM-0053 — Plan

A governed arrangement of Tasks, dependencies, resources, decision points, verification steps, and recovery conditions intended to realize an Objective.

Required distinction: A Plan is not proof that Actions are permitted, executed, or successful.

Type: Coordinated intended work | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0053` when it uses **Plan** with this exact governed meaning: A governed arrangement of Tasks, dependencies, resources, decision points, verification steps, and recovery conditions intended to realize an Objective.

*Counterexample: A dependent artifact uses **Plan** in a way that violates its required distinction: A Plan is not proof that Actions are permitted, executed, or successful.*

Relationships: HAL-REL-0017, HAL-REL-0018 | Lifecycle: None registered

Constraint: A Plan is not proof that Actions are permitted, executed, or successful.

HAL-TERM-0054 — Plan Graph

A dependency graph of intended work, decisions, resources, and verification gates.

Required distinction: It represents intended coordination, not completed reality.

Type: Planning representation | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0054` when it uses **Plan Graph** with this exact governed meaning: A dependency graph of intended work, decisions, resources, and verification gates.

*Counterexample: A dependent artifact uses **Plan Graph** in a way that violates its required distinction: It represents intended coordination, not completed reality.*

Relationships: None registered | Lifecycle: None registered

Constraint: It represents intended coordination, not completed reality.

HAL-TERM-0055 — Execution Graph

The governed graph of actual Attempts, Actions, dependencies, outcomes, and evidence for an execution instance.

Required distinction: It must not be silently substituted for the Plan Graph when explaining divergence.

Type: Runtime representation | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 12, 18, and 46 | Book II: Book II Chapters 06 and 07 | Book III: Book III Chapters 3, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0055` when it uses **Execution Graph** with this exact governed meaning: The governed graph of actual Attempts, Actions, dependencies, outcomes, and evidence for an execution instance.

*Counterexample: A dependent artifact uses **Execution Graph** in a way that violates its required distinction: It must not be silently substituted for the Plan Graph when explaining divergence.*

Relationships: None registered | Lifecycle: None registered

Constraint: It must not be silently substituted for the Plan Graph when explaining divergence.

HAL-TERM-0056 — Attention Object

A durable object representing a candidate matter for bounded attention, including source, salience, urgency, risk, relevance, context, disposition, and evidence.

Required distinction: Attention is not Authority, approval, or a promise to act.

Type: Prioritization record | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 54 | Book II: Book II Chapter 08 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0056` when it uses **Attention Object** with this exact governed meaning: A durable object representing a candidate matter for bounded attention, including source, salience, urgency, risk, relevance, context, disposition, and evidence.

*Counterexample: A dependent artifact uses **Attention Object** in a way that violates its required distinction: Attention is not Authority, approval, or a promise to act.*

Relationships: None registered | Lifecycle: None registered

Constraint: Attention is not Authority, approval, or a promise to act.

HAL-TERM-0057 — Decision Object

A durable record of a consequential decision including question, alternatives, authority context, evidence, uncertainty, judgment, rationale, selected disposition, and review conditions.

Required distinction: It records a decision and must not be used as a substitute for required Permission or execution evidence.

Type: Decision record | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 55 | Book II: Book II Chapter 09 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0057` when it uses **Decision Object** with this exact governed meaning: A durable record of a consequential decision including question, alternatives, authority context, evidence, uncertainty, judgment, rationale, selected disposition, and review conditions.

*Counterexample: A dependent artifact uses **Decision Object** in a way that violates its required distinction: It records a decision and must not be used as a substitute for required Permission or execution evidence.*

Relationships: HAL-REL-0019 | Lifecycle: None registered

Constraint: It records a decision and must not be used as a substitute for required Permission or execution evidence.

HAL-TERM-0058 — Judgment

The context-sensitive evaluation that weighs evidence, uncertainty, values, consequences, proportionality, and restraint to reach or recommend a decision.

Required distinction: Judgment must not silently invent authority or conceal unresolved uncertainty.

Type: Reasoned evaluation | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 55 | Book II: Book II Chapter 09 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0058` when it uses **Judgment** with this exact governed meaning: The context-sensitive evaluation that weighs evidence, uncertainty, values, consequences, proportionality, and restraint to reach or recommend a decision.

*Counterexample: A dependent artifact uses **Judgment** in a way that violates its required distinction: Judgment must not silently invent authority or conceal unresolved uncertainty.*

Relationships: HAL-REL-0019, HAL-REL-0036 | Lifecycle: None registered

Constraint: Judgment must not silently invent authority or conceal unresolved uncertainty.

HAL-TERM-0059 — Uncertainty

A represented limitation in knowledge, evidence, prediction, interpretation, or confidence relevant to a claim or decision.

Required distinction: Uncertainty is not failure; unrepresented material uncertainty is a defect.

Type: Epistemic condition | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 55 | Book II: Book II Chapter 09 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0059` when it uses **Uncertainty** with this exact governed meaning: A represented limitation in knowledge, evidence, prediction, interpretation, or confidence relevant to a claim or decision.

*Counterexample: A dependent artifact uses **Uncertainty** in a way that violates its required distinction: Uncertainty is not failure; unrepresented material uncertainty is a defect.*

Relationships: None registered | Lifecycle: None registered

Constraint: Uncertainty is not failure; unrepresented material uncertainty is a defect.

HAL-TERM-0060 — Outcome Object

A durable record linking intended outcome, observed result, affected parties, evidence, side effects, confidence, and evaluation.

Required distinction: It is not equivalent to an Event, metric sample, or optimistic status assertion.

Type: Outcome record | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 57 | Book II: Book II Chapter 32 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0060` when it uses **Outcome Object** with this exact governed meaning: A durable record linking intended outcome, observed result, affected parties, evidence, side effects, confidence, and evaluation.

*Counterexample: A dependent artifact uses **Outcome Object** in a way that violates its required distinction: It is not equivalent to an Event, metric sample, or optimistic status assertion.*

Relationships: HAL-REL-0020 | Lifecycle: None registered

Constraint: It is not equivalent to an Event, metric sample, or optimistic status assertion.

HAL-TERM-0061 — Success

A source-governed determination that relevant outcomes satisfy stated criteria without unacceptable constitutional, human, privacy, security, or reliability costs.

Required distinction: Task completion, activity volume, or a single metric does not by itself establish Success.

Type: Evaluated condition | Category: Cognition | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 57 | Book II: Book II Chapter 32 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0061` when it uses **Success** with this exact governed meaning: A source-governed determination that relevant outcomes satisfy stated criteria without unacceptable constitutional, human, privacy, security, or reliability costs.

*Counterexample: A dependent artifact uses **Success** in a way that violates its required distinction: Task completion, activity volume, or a single metric does not by itself establish Success.*

Relationships: None registered | Lifecycle: None registered

Constraint: Task completion, activity volume, or a single metric does not by itself establish Success.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Intent locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 6 — Capabilities, Providers, Actions, Transactions, and the Reality Boundary

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines abilities, implementations, governed execution, commit barriers, rollback, compensation, and separation of simulated from real effects. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I Decisions 10, 15, 16, 20, 23, 24, 35, 36, 44, and 50.

Book II: Book II Chapters 15, 16, 17, 22, 27, and 28.

Book III: Book III Chapters 03, 05, 06, and 07.

Normative semantic rules

21. A Capability MUST define an implementation-independent ability; Provider and Adapter records MUST identify implementations without redefining that ability.
22. Every real Action MUST cross an explicit Commit Barrier with applicable Authority, Permission, verification, and evidence.
23. Simulation, Digital Twin, and Shadow Execution MUST remain incapable of ungoverned real effects.
24. Rollback MUST be used only for truthful reversal; Compensation MUST name a new remedial Action.

Canonical term set

HAL-TERM-0062 — Capability

An abstract ability defined by outcomes, inputs, outputs, constraints, required authority and permission classes, risks, side effects, and evaluation criteria.

Required distinction: Capability does not identify an implementation and does not grant authority to use the ability.

Type: Implementation-independent contract | Category: Capability | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36 | Book II: Book II Chapter 15 | Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0062` when it uses **Capability** with this exact governed meaning: An abstract ability defined by outcomes, inputs, outputs, constraints, required authority and permission classes, risks, side effects, and evaluation criteria.

*Counterexample: A dependent artifact uses **Capability** in a way that violates its required distinction: Capability does not identify an implementation and does not grant authority to use the ability.*

Relationships: HAL-REL-0021, HAL-REL-0022 | Lifecycle: None registered

Constraint: Capability does not identify an implementation and does not grant authority to use the ability.

HAL-TERM-0063 — Capability Contract

The versioned specification of a Capability's semantic inputs, outputs, preconditions, effects, risks, authority requirements, evidence, and compatibility.

Required distinction: It is not a provider-specific API contract.

Type: Contract record | Category: Capability | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36 | Book II: Book II Chapter 15 | Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0063` when it uses **Capability Contract** with this exact governed meaning: The versioned specification of a Capability's semantic inputs, outputs, preconditions, effects, risks, authority requirements, evidence, and compatibility.

*Counterexample: A dependent artifact uses **Capability Contract** in a way that violates its required distinction: It is not a provider-specific API contract.*

Relationships: HAL-REL-0021 | Lifecycle: None registered

Constraint: It is not a provider-specific API contract.

HAL-TERM-0064 — Provider

A component, service, model, person, device, or external system that can fulfill a Capability under a declared contract and trust context.

Required distinction: Being able to perform work does not authorize the Provider to perform it.

Type: Implementation role | Category: Capability | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36 | Book II: Book II Chapter 15 | Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0064` when it uses **Provider** with this exact governed meaning: A component, service, model, person, device, or external system that can fulfill a Capability under a declared contract and trust context.

*Counterexample: A dependent artifact uses **Provider** in a way that violates its required distinction: Being able to perform work does not authorize the Provider to perform it.*

Relationships: HAL-REL-0022, HAL-REL-0023 | Lifecycle: None registered

Constraint: Being able to perform work does not authorize the Provider to perform it.

HAL-TERM-0065 — Adapter

A component that translates between a Capability Contract and a provider-specific interface while preserving authority, semantics, evidence, and failure behavior.

Required distinction: An Adapter must not smuggle provider semantics into the canonical Capability definition.

Type: Boundary component role | Category: Capability | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36 | Book II: Book II Chapter 15 | Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0065` when it uses **Adapter** with this exact governed meaning: A component that translates between a Capability Contract and a provider-specific interface while preserving authority, semantics, evidence, and failure behavior.

*Counterexample: A dependent artifact uses **Adapter** in a way that violates its required distinction: An Adapter must not smuggle provider semantics into the canonical Capability definition.*

Relationships: HAL-REL-0023 | Lifecycle: None registered

Constraint: An Adapter must not smuggle provider semantics into the canonical Capability definition.

HAL-TERM-0066 — Capability Registry

The governed catalog of Capabilities, versions, Providers, constraints, authority classes, health, and selection metadata.

Required distinction: Registration does not establish permission for use or trustworthiness in every domain.

Type: Authoritative registry | Category: Capability | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 10 and 36 | Book II: Book II Chapter 15 | Book III: Book III Chapters 3, 5, and 7

Example: A dependent artifact cites `HAL-TERM-0066` when it uses **Capability Registry** with this exact governed meaning: The governed catalog of Capabilities, versions, Providers, constraints, authority classes, health, and selection metadata.

*Counterexample: A dependent artifact uses **Capability Registry** in a way that violates its required distinction: Registration does not establish permission for use or trustworthiness in every domain.*

Relationships: None registered | Lifecycle: None registered

Constraint: Registration does not establish permission for use or trustworthiness in every domain.

HAL-TERM-0067 — Action

A governed attempt to produce an effect in authoritative state or the external world.

Required distinction: A read-only Query is not an Action; an Action is not proof of successful effect.

Type: State-changing attempt | Category: Action | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50 | Book II: Book II Chapter 16 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0067` when it uses **Action** with this exact governed meaning: A governed attempt to produce an effect in authoritative state or the external world.

*Counterexample: A dependent artifact uses **Action** in a way that violates its required distinction: A read-only Query is not an Action; an Action is not proof of successful effect.*

Relationships: HAL-REL-0024 | Lifecycle: None registered

Constraint: A read-only Query is not an Action; an Action is not proof of successful effect.

HAL-TERM-0068 — Attempt

One attributable execution effort for a Task, Action, or verification step with its own timing, context, result, and evidence.

Required distinction: A retry is a new Attempt even when it shares an idempotency key.

Type: Execution instance | Category: Action | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50 | Book II: Book II Chapter 16 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0068` when it uses **Attempt** with this exact governed meaning: One attributable execution effort for a Task, Action, or verification step with its own timing, context, result, and evidence.

*Counterexample: A dependent artifact uses **Attempt** in a way that violates its required distinction: A retry is a new Attempt even when it shares an idempotency key.*

Relationships: None registered | Lifecycle: None registered

Constraint: A retry is a new Attempt even when it shares an idempotency key.

HAL-TERM-0069 — Transaction

The durable coordination object for one or more Actions, including authorization, prepare, commit, result, evidence, rollback, and compensation states.

Required distinction: It is broader than a database transaction and must not imply atomic reversibility of external effects.

Type: Governed action lifecycle | Category: Action | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50 | Book II: Book II Chapter 16 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0069` when it uses **Transaction** with this exact governed meaning: The durable coordination object for one or more Actions, including authorization, prepare, commit, result, evidence, rollback, and compensation states.

*Counterexample: A dependent artifact uses **Transaction** in a way that violates its required distinction: It is broader than a database transaction and must not imply atomic reversibility of external effects.*

Relationships: HAL-REL-0024 | Lifecycle: HAL-TRANS-0008, HAL-TRANS-0009, HAL-TRANS-0010, HAL-TRANS-0011, HAL-TRANS-0012, HAL-TRANS-0013

Constraint: It is broader than a database transaction and must not imply atomic reversibility of external effects.

HAL-TERM-0070 — Commit Barrier

The explicit governed point after which a proposed change may create authoritative or real-world effects that cannot be treated as merely simulated or prepared.

Required distinction: Passing a Commit Barrier requires the applicable authority and verification; preparation alone must not cross it.

Type: Irreversibility gate | Category: Action | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50 | Book II: Book II Chapter 16 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0070` when it uses **Commit Barrier** with this exact governed meaning: The explicit governed point after which a proposed change may create authoritative or real-world effects that cannot be treated as merely simulated or prepared.

*Counterexample: A dependent artifact uses **Commit Barrier** in a way that violates its required distinction: Passing a Commit Barrier requires the applicable authority and verification; preparation alone must not cross it.*

Relationships: None registered | Lifecycle: None registered

Constraint: Passing a Commit Barrier requires the applicable authority and verification; preparation alone must not cross it.

HAL-TERM-0071 — Rollback

A controlled restoration of a prior recoverable state when the relevant effects are truthfully reversible.

Required distinction: Rollback must not claim to erase external or human effects that already occurred.

Type: Reversal operation | Category: Action | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50 | Book II: Book II Chapter 16 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0071` when it uses **Rollback** with this exact governed meaning: A controlled restoration of a prior recoverable state when the relevant effects are truthfully reversible.

*Counterexample: A dependent artifact uses **Rollback** in a way that violates its required distinction: Rollback must not claim to erase external or human effects that already occurred.*

Relationships: None registered | Lifecycle: None registered

Constraint: Rollback must not claim to erase external or human effects that already occurred.

HAL-TERM-0072 — Compensation

A new governed Action that mitigates, offsets, or repairs effects that cannot truthfully be undone.

Required distinction: Compensation is not Rollback and may require independent authority and evidence.

Type: Remedial operation | Category: Action | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 16, 35, and 50 | Book II: Book II Chapter 16 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0072` when it uses **Compensation** with this exact governed meaning: A new governed Action that mitigates, offsets, or repairs effects that cannot truthfully be undone.

*Counterexample: A dependent artifact uses **Compensation** in a way that violates its required distinction: Compensation is not Rollback and may require independent authority and evidence.*

Relationships: None registered | Lifecycle: None registered

Constraint: Compensation is not Rollback and may require independent authority and evidence.

HAL-TERM-0073 — Reality Boundary

The explicit separation among simulation, digital twin, shadow, test, canary, controlled-reality, production, recovery, and emergency contexts.

Required distinction: Non-reality authority, data, or effects must not leak across this boundary into reality.

Type: Governed environment boundary | Category: Action | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50 | Book II: Book II Chapter 17 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0073` when it uses **Reality Boundary** with this exact governed meaning: The explicit separation among simulation, digital twin, shadow, test, canary, controlled-reality, production, recovery, and emergency contexts.

*Counterexample: A dependent artifact uses **Reality Boundary** in a way that violates its required distinction: Non-reality authority, data, or effects must not leak across this boundary into reality.*

Relationships: HAL-REL-0025 | Lifecycle: None registered

Constraint: Non-reality authority, data, or effects must not leak across this boundary into reality.

HAL-TERM-0074 — Simulation

An execution environment whose effects are confined to modeled or synthetic state and cannot directly alter production or external reality.

Required distinction: High apparent fidelity does not make a Simulation production.

Type: Non-reality environment | Category: Verification | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50 | Book II: Book II Chapter 17 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0074` when it uses **Simulation** with this exact governed meaning: An execution environment whose effects are confined to modeled or synthetic state and cannot directly alter production or external reality.

*Counterexample: A dependent artifact uses **Simulation** in a way that violates its required distinction: High apparent fidelity does not make a Simulation production.*

Relationships: None registered | Lifecycle: None registered

Constraint: High apparent fidelity does not make a Simulation production.

HAL-TERM-0075 — Digital Twin

A governed model of selected real entities, relationships, state, and dynamics used to evaluate behavior without treating modeled effects as real effects.

Required distinction: A Digital Twin is bounded by declared fidelity and must not be mistaken for the represented reality.

Type: Modeled counterpart | Category: Verification | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50 | Book II: Book II Chapter 17 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0075` when it uses **Digital Twin** with this exact governed meaning: A governed model of selected real entities, relationships, state, and dynamics used to evaluate behavior without treating modeled effects as real effects.

*Counterexample: A dependent artifact uses **Digital Twin** in a way that violates its required distinction: A Digital Twin is bounded by declared fidelity and must not be mistaken for the represented reality.*

Relationships: None registered | Lifecycle: None registered

Constraint: A Digital Twin is bounded by declared fidelity and must not be mistaken for the represented reality.

HAL-TERM-0076 — Shadow Execution

Execution using live or representative inputs while preventing proposed outputs from producing authoritative or external effects.

Required distinction: Shadow results do not authorize promotion without the required review and certification.

Type: Non-committing execution mode | Category: Verification | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50 | Book II: Book II Chapter 17 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0076` when it uses **Shadow Execution** with this exact governed meaning: Execution using live or representative inputs while preventing proposed outputs from producing authoritative or external effects.

*Counterexample: A dependent artifact uses **Shadow Execution** in a way that violates its required distinction: Shadow results do not authorize promotion without the required review and certification.*

Relationships: None registered | Lifecycle: None registered

Constraint: Shadow results do not authorize promotion without the required review and certification.

HAL-TERM-0077 — Canary

A deliberately constrained real execution stage used to accumulate evidence before broader adoption.

Required distinction: A Canary crosses the Reality Boundary and therefore requires real authority, containment, and rollback or compensation.

Type: Limited reality stage | Category: Verification | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 50 | Book II: Book II Chapter 17 | Book III: Book III Chapters 3, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0077` when it uses **Canary** with this exact governed meaning: A deliberately constrained real execution stage used to accumulate evidence before broader adoption.

*Counterexample: A dependent artifact uses **Canary** in a way that violates its required distinction: A Canary crosses the Reality Boundary and therefore requires real authority, containment, and rollback or compensation.*

Relationships: HAL-REL-0025 | Lifecycle: None registered

Constraint: A Canary crosses the Reality Boundary and therefore requires real authority, containment, and rollback or compensation.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Capability locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 7 — Evidence, Trust, Verification, Assurance, and Certification

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines the evidence admission boundary, claims, provenance, custody, trust assessment, verification, assurance cases, and certification. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I Articles II and XI; Decisions 22, 26, 34, 35, 40, 42, 43, 50, and 56.

Book II: Book II Chapters 17, 18, 25, and 35.

Book III: Book III Chapters 04, 06, 08, and 09.

Normative semantic rules

25. An Evidence Object MUST be immutable after admission; correction MUST occur through linked superseding, challenging, or explanatory objects.
26. An Evidence Candidate or Audit Record MUST NOT be represented as an Evidence Object before governed admission.
27. Verification MUST identify claims, criteria, methods, environment, evidence, uncertainty, and reproducibility.
28. Certification MUST be scoped, time-bounded, attributable, evidence-based, suspendable, and revocable.

Canonical term set

HAL-TERM-0078 — Evidence Candidate

An observation, telemetry item, document, Audit Record, claim, or other integrity-protected input proposed for admission as an Evidence Object.

Required distinction: It is not authoritative Evidence until accepted through the Evidence Service's governed admission and custody process.

Type: Pre-admission record | Category: Evidence | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26 | Book II: Book II Chapters 18 and 25 | Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0078` when it uses **Evidence Candidate** with this exact governed meaning: An observation, telemetry item, document, Audit Record, claim, or other integrity-protected input proposed for admission as an Evidence Object.

*Counterexample: A dependent artifact uses **Evidence Candidate** in a way that violates its required distinction: It is not authoritative Evidence until accepted through the Evidence Service's governed admission and custody process.*

Relationships: HAL-REL-0026 | Lifecycle: HAL-TRANS-0014

Constraint: It is not authoritative Evidence until accepted through the Evidence Service's governed admission and custody process.

HAL-TERM-0079 — Evidence Object

An immutable object admitted and governed by the authoritative Evidence Service that records provenance, custody, source identity, observation or claim content, time, signatures, verification state, confidence, domain, and expiration metadata.

Required distinction: An Evidence Object must not be mutated to revise a conclusion; later objects supersede or challenge it.

Type: Immutable provenance-bearing record | Category: Evidence | Status: Approved | Aliases: Evidence

Source basis: Direct source normalization

Book I: Book I Decision 26 | Book II: Book II Chapters 18 and 25 | Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0079` when it uses **Evidence Object** with this exact governed meaning: An immutable object admitted and governed by the authoritative Evidence Service that records provenance, custody, source identity, observation or claim content, time, signatures, verification state, confidence, domain, and expiration metadata.

*Counterexample: A dependent artifact uses **Evidence Object** in a way that violates its required distinction: An Evidence Object must not be mutated to revise a conclusion; later objects supersede or challenge it.*

Relationships: HAL-REL-0026, HAL-REL-0027, HAL-REL-0028, HAL-REL-0053 | Lifecycle: None registered

Constraint: An Evidence Object must not be mutated to revise a conclusion; later objects supersede or challenge it.

HAL-TERM-0080 — Audit Record

An append-only record of a protected action, authorization, decision, access, or change owned by the applicable audit domain.

Required distinction: An Audit Record may produce an Evidence Candidate but is not automatically a general Evidence Object.

Type: Protected accountability record | Category: Evidence | Status: Approved | Aliases: audit log entry

Source basis: Direct source normalization

Book I: Book I Decision 26 | Book II: Book II Chapters 18 and 25 | Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0080` when it uses **Audit Record** with this exact governed meaning: An append-only record of a protected action, authorization, decision, access, or change owned by the applicable audit domain.

*Counterexample: A dependent artifact uses **Audit Record** in a way that violates its required distinction: An Audit Record may produce an Evidence Candidate but is not automatically a general Evidence Object.*

Relationships: None registered | Lifecycle: None registered

Constraint: An Audit Record may produce an Evidence Candidate but is not automatically a general Evidence Object.

HAL-TERM-0081 — Provenance

The attributable origin, derivation, transformation, and custody history of an artifact, datum, claim, model, decision, or Evidence Object.

Required distinction: A source label without derivation and custody context is incomplete provenance.

Type: Origin history | Category: Evidence | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26 | Book II: Book II Chapters 18 and 25 | Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0081` when it uses **Provenance** with this exact governed meaning: The attributable origin, derivation, transformation, and custody history of an artifact, datum, claim, model, decision, or Evidence Object.

*Counterexample: A dependent artifact uses **Provenance** in a way that violates its required distinction: A source label without derivation and custody context is incomplete provenance.*

Relationships: None registered | Lifecycle: None registered

Constraint: A source label without derivation and custody context is incomplete provenance.

HAL-TERM-0082 — Chain of Custody

The ordered, attributable record of possession, control, transfer, and integrity protection for evidence or sensitive artifacts.

Required distinction: It does not establish truth; it establishes accountable handling.

Type: Integrity history | Category: Evidence | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26 | Book II: Book II Chapters 18 and 25 | Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0082` when it uses **Chain of Custody** with this exact governed meaning: The ordered, attributable record of possession, control, transfer, and integrity protection for evidence or sensitive artifacts.

*Counterexample: A dependent artifact uses **Chain of Custody** in a way that violates its required distinction: It does not establish truth; it establishes accountable handling.*

Relationships: None registered | Lifecycle: None registered

Constraint: It does not establish truth; it establishes accountable handling.

HAL-TERM-0083 — Evidence Graph

A graph connecting Claims, Evidence Objects, sources, derivations, supporting or opposing relations, confidence, and conclusions.

Required distinction: Graph connectivity does not make all linked material equally authoritative or trustworthy.

Type: Evidence relationship model | Category: Evidence | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26 | Book II: Book II Chapters 18 and 25 | Book III: Book III Chapters 4, 5, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0083` when it uses **Evidence Graph** with this exact governed meaning: A graph connecting Claims, Evidence Objects, sources, derivations, supporting or opposing relations, confidence, and conclusions.

*Counterexample: A dependent artifact uses **Evidence Graph** in a way that violates its required distinction: Graph connectivity does not make all linked material equally authoritative or trustworthy.*

Relationships: HAL-REL-0028 | Lifecycle: None registered

Constraint: Graph connectivity does not make all linked material equally authoritative or trustworthy.

HAL-TERM-0084 — Verification

A reproducible, risk-scaled process that evaluates a Claim, invariant, behavior, artifact, or Outcome against explicit criteria.

Required distinction: Verification produces evidence; it is not the same as certification or operational approval.

Type: Evidence-producing evaluation | Category: Verification | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56 | Book II: Book II Chapters 17 and 35 | Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0084` when it uses **Verification** with this exact governed meaning: A reproducible, risk-scaled process that evaluates a Claim, invariant, behavior, artifact, or Outcome against explicit criteria.

*Counterexample: A dependent artifact uses **Verification** in a way that violates its required distinction: Verification produces evidence; it is not the same as certification or operational approval.*

Relationships: HAL-REL-0029 | Lifecycle: None registered

Constraint: Verification produces evidence; it is not the same as certification or operational approval.

HAL-TERM-0085 — Verification Plan

A governed specification of claims, risks, methods, environments, data, success criteria, independence, and required evidence for Verification.

Required distinction: A test list without mapped claims and criteria is not a complete Verification Plan.

Type: Planning record | Category: Verification | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56 | Book II: Book II Chapters 17 and 35 | Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0085` when it uses **Verification Plan** with this exact governed meaning: A governed specification of claims, risks, methods, environments, data, success criteria, independence, and required evidence for Verification.

*Counterexample: A dependent artifact uses **Verification Plan** in a way that violates its required distinction: A test list without mapped claims and criteria is not a complete Verification Plan.*

Relationships: None registered | Lifecycle: None registered

Constraint: A test list without mapped claims and criteria is not a complete Verification Plan.

HAL-TERM-0086 — Assurance Case

A structured, reviewable argument connecting scoped claims to reasoning and sufficient supporting evidence.

Required distinction: An Assurance Case must expose assumptions, defeaters, uncertainty, and evidence gaps.

Type: Structured argument | Category: Verification | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56 | Book II: Book II Chapters 17 and 35 | Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0086` when it uses **Assurance Case** with this exact governed meaning: A structured, reviewable argument connecting scoped claims to reasoning and sufficient supporting evidence.

*Counterexample: A dependent artifact uses **Assurance Case** in a way that violates its required distinction: An Assurance Case must expose assumptions, defeaters, uncertainty, and evidence gaps.*

Relationships: HAL-REL-0030, HAL-REL-0031 | Lifecycle: None registered

Constraint: An Assurance Case must expose assumptions, defeaters, uncertainty, and evidence gaps.

HAL-TERM-0087 — Certification

A scoped, time-bounded, evidence-based determination by an authorized certifier that specified conformance claims are satisfied.

Required distinction: Certification is not permanent, universal, or self-issued by the artifact being certified.

Type: Governed assurance decision | Category: Verification | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56 | Book II: Book II Chapters 17 and 35 | Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0087` when it uses **Certification** with this exact governed meaning: A scoped, time-bounded, evidence-based determination by an authorized certifier that specified conformance claims are satisfied.

*Counterexample: A dependent artifact uses **Certification** in a way that violates its required distinction: Certification is not permanent, universal, or self-issued by the artifact being certified.*

Relationships: HAL-REL-0031 | Lifecycle: HAL-TRANS-0015, HAL-TRANS-0016, HAL-TRANS-0017

Constraint: Certification is not permanent, universal, or self-issued by the artifact being certified.

HAL-TERM-0088 — Conformance

The evidenced condition of satisfying identified requirements from identified authoritative sources within a declared scope and version.

Required distinction: Conformance is never implied merely by compatibility, successful execution, or absence of known defects.

Type: Evaluated relation | Category: Verification | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56 | Book II: Book II Chapters 17 and 35 | Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0088` when it uses **Conformance** with this exact governed meaning: The evidenced condition of satisfying identified requirements from identified authoritative sources within a declared scope and version.

*Counterexample: A dependent artifact uses **Conformance** in a way that violates its required distinction: Conformance is never implied merely by compatibility, successful execution, or absence of known defects.*

Relationships: None registered | Lifecycle: None registered

Constraint: Conformance is never implied merely by compatibility, successful execution, or absence of known defects.

HAL-TERM-0089 — Confidence

A bounded assessment of support for a Claim or prediction, expressed with method, scope, evidence basis, uncertainty, and time sensitivity.

Required distinction: Confidence is not probability unless a defined model justifies that interpretation.

Type: Calibrated assessment | Category: Trust | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 43, 50, and 56 | Book II: Book II Chapters 17 and 35 | Book III: Book III Chapters 6, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0089` when it uses **Confidence** with this exact governed meaning: A bounded assessment of support for a Claim or prediction, expressed with method, scope, evidence basis, uncertainty, and time sensitivity.

*Counterexample: A dependent artifact uses **Confidence** in a way that violates its required distinction: Confidence is not probability unless a defined model justifies that interpretation.*

Relationships: None registered | Lifecycle: None registered

Constraint: Confidence is not probability unless a defined model justifies that interpretation.

HAL-TERM-0165 — Evidence Service

The Book II authoritative service that admits and governs Evidence Objects and owns their custody, signatures, provenance bindings, and verification state.

Required distinction: Observability, audit, and source systems may produce Evidence Candidates or records but cannot independently admit Evidence Objects or mutate evidentiary meaning.

Type: Architectural component class | Category: Evidence | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 26 | Book II: Book II Chapters 18 and 25 | Book III: Book III Chapters 4, 5, 6, and 8

Example: A telemetry record enters as an Evidence Candidate; the Evidence Service validates provenance and custody before admitting a new immutable Evidence Object.

Counterexample: A logging or observability service labels its mutable record an Evidence Object without the governed admission process.

Relationships: HAL-REL-0053 | Lifecycle: None registered

Constraint: Observability, audit, and source systems may produce Evidence Candidates or records but cannot independently admit Evidence Objects or mutate evidentiary meaning.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Evidence Candidate locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 8 — Experience, Memory, Knowledge, Learning, Patterns, and Wisdom

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Separates records of occurrence, retained experience, contextualized knowledge, learned patterns, and evidence-bounded wisdom. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I learning, wisdom, evidence, uncertainty, restraint, and continuity.

Book II: Book II Chapters 10, 11, 12, 13, and 30.

Book III: Book III Chapters 03, 04, 05, 06, and 08.

Normative semantic rules

29. Experience, Memory, Knowledge, Pattern, and Wisdom MUST remain distinct by provenance, epistemic status, purpose, and lifecycle.
30. Learning MUST NOT silently modify constitutional meaning, Authority, protected behavior, or production state.
31. Patterns MUST state supporting evidence, domain, confidence, limitations, exceptions, and review horizon.
32. Wisdom MAY inform Judgment but MUST NOT create Permission or Authority.

Canonical term set

HAL-TERM-0090 — Experience

A governed representation of what HAL perceived, attempted, decided, experienced, and observed, with context, outcomes, provenance, and privacy controls.

Required distinction: Experience is not automatically Knowledge, a Pattern, or Wisdom.

Type: Retained occurrence record | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 30 and 53 | Book II: Book II Chapter 12 | Book III: Book III Chapters 4, 5, and 6

Example: A dependent artifact cites `HAL-TERM-0090` when it uses **Experience** with this exact governed meaning: A governed representation of what HAL perceived, attempted, decided, experienced, and observed, with context, outcomes, provenance, and privacy controls.

*Counterexample: A dependent artifact uses **Experience** in a way that violates its required distinction: Experience is not automatically Knowledge, a Pattern, or Wisdom.*

Relationships: HAL-REL-0032, HAL-REL-0035 | Lifecycle: None registered

Constraint: Experience is not automatically Knowledge, a Pattern, or Wisdom.

HAL-TERM-0091 — Experience Ledger

The append-oriented governed store of Experience records and their provenance, correction, retention, and access metadata.

Required distinction: It is not a general-purpose mutable memory store.

Type: Authoritative ledger | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 30 and 53 | Book II: Book II Chapter 12 | Book III: Book III Chapters 4, 5, and 6

Example: A dependent artifact cites `HAL-TERM-0091` when it uses **Experience Ledger** with this exact governed meaning: The append-oriented governed store of Experience records and their provenance, correction, retention, and access metadata.

*Counterexample: A dependent artifact uses **Experience Ledger** in a way that violates its required distinction: It is not a general-purpose mutable memory store.*

Relationships: HAL-REL-0032 | Lifecycle: None registered

Constraint: It is not a general-purpose mutable memory store.

HAL-TERM-0092 — Memory

A retained representation available for later contextual retrieval under authority, privacy, relevance, and lifecycle rules.

Required distinction: Memory is broader than Experience and is not necessarily authoritative Knowledge.

Type: Retrievable retained representation | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 30 and 53 | Book II: Book II Chapter 12 | Book III: Book III Chapters 4, 5, and 6

Example: A dependent artifact cites `HAL-TERM-0092` when it uses **Memory** with this exact governed meaning: A retained representation available for later contextual retrieval under authority, privacy, relevance, and lifecycle rules.

*Counterexample: A dependent artifact uses **Memory** in a way that violates its required distinction: Memory is broader than Experience and is not necessarily authoritative Knowledge.*

Relationships: HAL-REL-0033 | Lifecycle: None registered

Constraint: Memory is broader than Experience and is not necessarily authoritative Knowledge.

HAL-TERM-0093 — Memory Graph

A governed graph linking retained representations by context, entity, time, causation, similarity, and relevance.

Required distinction: Association must not be treated as proof of causation or truth.

Type: Associative representation | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 30 and 53 | Book II: Book II Chapter 12 | Book III: Book III Chapters 4, 5, and 6

Example: A dependent artifact cites `HAL-TERM-0093` when it uses **Memory Graph** with this exact governed meaning: A governed graph linking retained representations by context, entity, time, causation, similarity, and relevance.

*Counterexample: A dependent artifact uses **Memory Graph** in a way that violates its required distinction: Association must not be treated as proof of causation or truth.*

Relationships: HAL-REL-0033 | Lifecycle: None registered

Constraint: Association must not be treated as proof of causation or truth.

HAL-TERM-0094 — Knowledge

A governed representation whose claims, provenance, validity scope, confidence, and supporting evidence are sufficient for its declared use.

Required distinction: Stored information or model output is not automatically Knowledge.

Type: Contextualized warranted representation | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 30 and 52 | Book II: Book II Chapter 10 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0094` when it uses **Knowledge** with this exact governed meaning: A governed representation whose claims, provenance, validity scope, confidence, and supporting evidence are sufficient for its declared use.

*Counterexample: A dependent artifact uses **Knowledge** in a way that violates its required distinction: Stored information or model output is not automatically Knowledge.*

Relationships: HAL-REL-0034 | Lifecycle: None registered

Constraint: Stored information or model output is not automatically Knowledge.

HAL-TERM-0095 — Knowledge Graph

A governed graph of entities, concepts, relationships, Claims, sources, validity, and provenance used for contextual reasoning and retrieval.

Required distinction: Graph membership does not erase source authority, uncertainty, or temporal scope.

Type: Knowledge relationship model | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 30 and 52 | Book II: Book II Chapter 10 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0095` when it uses **Knowledge Graph** with this exact governed meaning: A governed graph of entities, concepts, relationships, Claims, sources, validity, and provenance used for contextual reasoning and retrieval.

*Counterexample: A dependent artifact uses **Knowledge Graph** in a way that violates its required distinction: Graph membership does not erase source authority, uncertainty, or temporal scope.*

Relationships: HAL-REL-0034 | Lifecycle: None registered

Constraint: Graph membership does not erase source authority, uncertainty, or temporal scope.

HAL-TERM-0096 — Pattern

A reproducibly supported regularity across Experiences or observations with stated domain, evidence, confidence, limits, and exceptions.

Required distinction: A repeated coincidence or one-off anecdote is not a Pattern.

Type: Learned regularity | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 31 and 53 | Book II: Book II Chapter 11 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0096` when it uses **Pattern** with this exact governed meaning: A reproducibly supported regularity across Experiences or observations with stated domain, evidence, confidence, limits, and exceptions.

*Counterexample: A dependent artifact uses **Pattern** in a way that violates its required distinction: A repeated coincidence or one-off anecdote is not a Pattern.*

Relationships: HAL-REL-0035 | Lifecycle: None registered

Constraint: A repeated coincidence or one-off anecdote is not a Pattern.

HAL-TERM-0097 — Learning

The evidence-bounded process by which HAL updates representations, Patterns, policies, or behavior within authorized scope.

Required distinction: Learning must not silently alter constitutional meaning, authority, or protected production behavior.

Type: Governed update process | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 31 and 53 | Book II: Book II Chapter 11 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0097` when it uses **Learning** with this exact governed meaning: The evidence-bounded process by which HAL updates representations, Patterns, policies, or behavior within authorized scope.

*Counterexample: A dependent artifact uses **Learning** in a way that violates its required distinction: Learning must not silently alter constitutional meaning, authority, or protected production behavior.*

Relationships: None registered | Lifecycle: None registered

Constraint: Learning must not silently alter constitutional meaning, authority, or protected production behavior.

HAL-TERM-0098 — Learning Ledger

The governed record of learning proposals, evidence, evaluation, approval, applied changes, monitoring, rollback, and outcomes.

Required distinction: It is not permission for unrestricted self-modification.

Type: Change ledger | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 31 and 53 | Book II: Book II Chapter 11 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0098` when it uses **Learning Ledger** with this exact governed meaning: The governed record of learning proposals, evidence, evaluation, approval, applied changes, monitoring, rollback, and outcomes.

*Counterexample: A dependent artifact uses **Learning Ledger** in a way that violates its required distinction: It is not permission for unrestricted self-modification.*

Relationships: None registered | Lifecycle: None registered

Constraint: It is not permission for unrestricted self-modification.

HAL-TERM-0099 — Wisdom

A durable, revisable synthesis of Experience, Patterns, values, consequences, uncertainty, and restraint used to inform Judgment.

Required distinction: Wisdom informs decisions but does not create authority or replace current evidence.

Type: Evidence-bounded judgment resource | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 31 and 53 | Book II: Book II Chapter 11 | Book III: Book III Chapters 4, 6, and 8

Example: A dependent artifact cites `HAL-TERM-0099` when it uses **Wisdom** with this exact governed meaning: A durable, revisable synthesis of Experience, Patterns, values, consequences, uncertainty, and restraint used to inform Judgment.

*Counterexample: A dependent artifact uses **Wisdom** in a way that violates its required distinction: Wisdom informs decisions but does not create authority or replace current evidence.*

Relationships: HAL-REL-0036 | Lifecycle: None registered

Constraint: Wisdom informs decisions but does not create authority or replace current evidence.

HAL-TERM-0100 — Self Model

HAL's evidence-linked representation of its current capabilities, limits, state, dependencies, uncertainty, and identity continuity.

Required distinction: It is descriptive and must not become a self-authorizing source.

Type: Governed self-representation | Category: Knowledge | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 28 and 51 | Book II: Book II Chapter 30 | Book III: Book III Chapters 4 and 8

Example: A dependent artifact cites `HAL-TERM-0100` when it uses **Self Model** with this exact governed meaning: HAL's evidence-linked representation of its current capabilities, limits, state, dependencies, uncertainty, and identity continuity.

*Counterexample: A dependent artifact uses **Self Model** in a way that violates its required distinction: It is descriptive and must not become a self-authorizing source.*

Relationships: None registered | Lifecycle: None registered

Constraint: It is descriptive and must not become a self-authorizing source.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Experience locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 9 — State, Time, Events, Messaging, Coordination, and Persistence

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines authoritative state, projections, commands, queries, events, ordering, time, idempotency, messaging, and durable persistence. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I evidence, continuity, reversibility, accountability, and failure containment.

Book II: Book II Chapters 13, 22, 23, 24, 25, 27, and 28.

Book III: Book III Chapters 03, 04, 06, and 07.

Normative semantic rules

33. Commands MUST request possible change; Events MUST state completed facts; Queries MUST NOT intentionally mutate authoritative state.
34. Each authoritative state domain MUST identify one mutation owner; projections, caches, and replicas MUST remain explicitly derived.
35. Ordering claims MUST state scope and mechanism; Wall-Clock Time MUST NOT be used as sole proof of distributed causality.
36. Idempotency claims MUST state operation, key scope, retention horizon, result semantics, and external-effect limitations.

Canonical term set

HAL-TERM-0101 — Command

A request addressed to an authoritative owner to evaluate and, if allowed, perform a state-changing operation.

Required distinction: A Command is not evidence that the operation was authorized, committed, or completed.

Type: Intent message | Category: Distributed systems | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40 | Book II: Book II Chapters 22 and 23 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0101` when it uses **Command** with this exact governed meaning: A request addressed to an authoritative owner to evaluate and, if allowed, perform a state-changing operation.

*Counterexample: A dependent artifact uses **Command** in a way that violates its required distinction: A Command is not evidence that the operation was authorized, committed, or completed.*

Relationships: HAL-REL-0037, HAL-REL-0040 | Lifecycle: None registered

Constraint: A Command is not evidence that the operation was authorized, committed, or completed.

HAL-TERM-0102 — Query

A request for information that must not intentionally mutate authoritative state or external reality.

Required distinction: Telemetry side effects must remain non-authoritative and must not turn a Query into a hidden Command.

Type: Read request | Category: Distributed systems | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40 | Book II: Book II Chapters 22 and 23 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0102` when it uses **Query** with this exact governed meaning: A request for information that must not intentionally mutate authoritative state or external reality.

*Counterexample: A dependent artifact uses **Query** in a way that violates its required distinction: Telemetry side effects must remain non-authoritative and must not turn a Query into a hidden Command.*

Relationships: None registered | Lifecycle: None registered

Constraint: Telemetry side effects must remain non-authoritative and must not turn a Query into a hidden Command.

HAL-TERM-0103 — Event

An immutable message stating that a defined fact or state transition has completed in its owning domain.

Required distinction: An Event is not a request, intention, or mutable current-state record.

Type: Completed-fact record | Category: Distributed systems | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40 | Book II: Book II Chapters 22 and 23 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0103` when it uses **Event** with this exact governed meaning: An immutable message stating that a defined fact or state transition has completed in its owning domain.

*Counterexample: A dependent artifact uses **Event** in a way that violates its required distinction: An Event is not a request, intention, or mutable current-state record.*

Relationships: HAL-REL-0037, HAL-REL-0039 | Lifecycle: None registered

Constraint: An Event is not a request, intention, or mutable current-state record.

HAL-TERM-0104 — Event Journal

The ordered durable record of Events for an owning domain, with identity, sequence, provenance, and integrity controls.

Required distinction: Global total order must not be inferred when only per-stream ordering exists.

Type: Durable event store | Category: Distributed systems | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40 | Book II: Book II Chapters 22 and 23 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0104` when it uses **Event Journal** with this exact governed meaning: The ordered durable record of Events for an owning domain, with identity, sequence, provenance, and integrity controls.

*Counterexample: A dependent artifact uses **Event Journal** in a way that violates its required distinction: Global total order must not be inferred when only per-stream ordering exists.*

Relationships: HAL-REL-0038 | Lifecycle: None registered

Constraint: Global total order must not be inferred when only per-stream ordering exists.

HAL-TERM-0105 — Message Envelope

The governed wrapper carrying message identity, schema version, source, destination, time, correlation, causation, authority context, classification, and integrity metadata.

Required distinction: The envelope does not change the semantic type of its payload.

Type: Transport record | Category: Distributed systems | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40 | Book II: Book II Chapters 22 and 23 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0105` when it uses **Message Envelope** with this exact governed meaning: The governed wrapper carrying message identity, schema version, source, destination, time, correlation, causation, authority context, classification, and integrity metadata.

*Counterexample: A dependent artifact uses **Message Envelope** in a way that violates its required distinction: The envelope does not change the semantic type of its payload.*

Relationships: HAL-REL-0040 | Lifecycle: None registered

Constraint: The envelope does not change the semantic type of its payload.

HAL-TERM-0106 — Idempotency Key

A scoped stable value allowing a receiver to recognize equivalent retry Attempts for one declared operation and result horizon.

Required distinction: It does not make non-idempotent external effects reversible or globally exactly-once.

Type: Deduplication value | Category: Distributed systems | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40 | Book II: Book II Chapters 22 and 23 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0106` when it uses **Idempotency Key** with this exact governed meaning: A scoped stable value allowing a receiver to recognize equivalent retry Attempts for one declared operation and result horizon.

*Counterexample: A dependent artifact uses **Idempotency Key** in a way that violates its required distinction: It does not make non-idempotent external effects reversible or globally exactly-once.*

Relationships: None registered | Lifecycle: None registered

Constraint: It does not make non-idempotent external effects reversible or globally exactly-once.

HAL-TERM-0107 — Correlation Identifier

A value linking related work, messages, evidence, and telemetry across a bounded flow.

Required distinction: Correlation does not prove causation.

Type: Trace reference | Category: Distributed systems | Status: Approved | Aliases: correlation ID

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40 | Book II: Book II Chapters 22 and 23 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0107` when it uses **Correlation Identifier** with this exact governed meaning: A value linking related work, messages, evidence, and telemetry across a bounded flow.

*Counterexample: A dependent artifact uses **Correlation Identifier** in a way that violates its required distinction: Correlation does not prove causation.*

Relationships: None registered | Lifecycle: None registered

Constraint: Correlation does not prove causation.

HAL-TERM-0108 — Causation Identifier

A value identifying the immediate initiating message, decision, or event for a derived operation.

Required distinction: It records declared lineage and must not be used as sole proof of real-world causality.

Type: Causal reference | Category: Distributed systems | Status: Approved | Aliases: causation ID

Source basis: Direct source normalization

Book I: Book I Decisions 22, 29, and 40 | Book II: Book II Chapters 22 and 23 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0108` when it uses **Causation Identifier** with this exact governed meaning: A value identifying the immediate initiating message, decision, or event for a derived operation.

*Counterexample: A dependent artifact uses **Causation Identifier** in a way that violates its required distinction: It records declared lineage and must not be used as sole proof of real-world causality.*

Relationships: None registered | Lifecycle: None registered

Constraint: It records declared lineage and must not be used as sole proof of real-world causality.

HAL-TERM-0109 — Authoritative State

State whose mutation and truth are governed by the designated owning domain.

Required distinction: A cache, replica, projection, search index, or local aggregate is not authoritative unless explicitly designated.

Type: Owned state | Category: Distributed systems | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40 | Book II: Book II Chapters 23 and 24 | Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0109` when it uses **Authoritative State** with this exact governed meaning: State whose mutation and truth are governed by the designated owning domain.

*Counterexample: A dependent artifact uses **Authoritative State** in a way that violates its required distinction: A cache, replica, projection, search index, or local aggregate is not authoritative unless explicitly designated.*

Relationships: None registered | Lifecycle: None registered

Constraint: A cache, replica, projection, search index, or local aggregate is not authoritative unless explicitly designated.

HAL-TERM-0110 — Projection

A read-optimized representation derived from authoritative records or Events for a declared purpose.

Required distinction: A Projection may be stale and must not silently accept authoritative mutations.

Type: Derived state | Category: Distributed systems | Status: Approved | Aliases: read model

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40 | Book II: Book II Chapters 23 and 24 | Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0110` when it uses **Projection** with this exact governed meaning: A read-optimized representation derived from authoritative records or Events for a declared purpose.

*Counterexample: A dependent artifact uses **Projection** in a way that violates its required distinction: A Projection may be stale and must not silently accept authoritative mutations.*

Relationships: HAL-REL-0038 | Lifecycle: None registered

Constraint: A Projection may be stale and must not silently accept authoritative mutations.

HAL-TERM-0111 — Cache

A replaceable performance-oriented copy whose loss does not destroy authoritative truth.

Required distinction: A Cache must not become the only copy of required state or evidence.

Type: Disposable derived state | Category: Distributed systems | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40 | Book II: Book II Chapters 23 and 24 | Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0111` when it uses **Cache** with this exact governed meaning: A replaceable performance-oriented copy whose loss does not destroy authoritative truth.

Counterexample: A dependent artifact uses **Cache** in a way that violates its required distinction: A Cache must not become the only copy of required state or evidence.

Relationships: None registered | Lifecycle: None registered

Constraint: A Cache must not become the only copy of required state or evidence.

HAL-TERM-0112 — Replica

A governed copy maintained from an authoritative source under declared consistency, lag, and failover rules.

Required distinction: Replication alone does not transfer semantic ownership.

Type: Replicated state | Category: Distributed systems | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40 | Book II: Book II Chapters 23 and 24 | Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0112` when it uses **Replica** with this exact governed meaning: A governed copy maintained from an authoritative source under declared consistency, lag, and failover rules.

Counterexample: A dependent artifact uses **Replica** in a way that violates its required distinction: Replication alone does not transfer semantic ownership.

Relationships: None registered | Lifecycle: None registered

Constraint: Replication alone does not transfer semantic ownership.

HAL-TERM-0113 — Transactional Outbox

A durable pattern that records an authoritative state change and its pending Event publication in one local commit boundary.

Required distinction: It reduces dual-write failure but does not guarantee global exactly-once processing.

Type: Publication pattern | Category: Distributed systems | Status: Approved | Aliases: outbox

Source basis: Direct source normalization

Book I: Book I Decisions 4, 22, 35, 38, and 40 | Book II: Book II Chapters 23 and 24 | Book III: Book III Chapters 3, 4, and 7

Example: A dependent artifact cites `HAL-TERM-0113` when it uses **Transactional Outbox** with this exact governed meaning: A durable pattern that records an authoritative state change and its pending Event publication in one local commit boundary.

Counterexample: A dependent artifact uses **Transactional Outbox** in a way that violates its required distinction: It reduces dual-write failure but does not guarantee global exactly-once processing.

Relationships: HAL-REL-0039 | Lifecycle: None registered

Constraint: It reduces dual-write failure but does not guarantee global exactly-once processing.

HAL-TERM-0114 — Logical Time

A non-wall-clock ordering value used to represent causal or domain sequence relationships.

Required distinction: Logical Time must not be presented as real elapsed or calendar time.

Type: Ordering construct | Category: Temporal | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 44 | Book II: Book II Chapter 13 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0114` when it uses **Logical Time** with this exact governed meaning: A non-wall-clock ordering value used to represent causal or domain sequence relationships.

*Counterexample: A dependent artifact uses **Logical Time** in a way that violates its required distinction: Logical Time must not be presented as real elapsed or calendar time.*

Relationships: None registered | Lifecycle: None registered

Constraint: Logical Time must not be presented as real elapsed or calendar time.

HAL-TERM-0115 — Wall-Clock Time

A timestamp from a declared clock source, precision, and synchronization context.

Required distinction: Wall-Clock Time alone must not establish distributed causal order.

Type: Temporal observation | Category: Temporal | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 44 | Book II: Book II Chapter 13 | Book III: Book III Chapters 3, 4, and 6

Example: A dependent artifact cites `HAL-TERM-0115` when it uses **Wall-Clock Time** with this exact governed meaning: A timestamp from a declared clock source, precision, and synchronization context.

*Counterexample: A dependent artifact uses **Wall-Clock Time** in a way that violates its required distinction: Wall-Clock Time alone must not establish distributed causal order.*

Relationships: None registered | Lifecycle: None registered

Constraint: Wall-Clock Time alone must not establish distributed causal order.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Command locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 10 — Privacy, Security, Trust Domains, Treaties, and the Constitutional Firewall

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines information classification, privacy purpose, secrets, credentials, trust boundaries, external domains, Treaties, and governed exchange. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I privacy, dignity, sovereignty, external trust, and protected authority.

Book II: Book II Chapters 18, 19, 20, 21, 25, and 26.

Book III: Book III Chapters 02, 04, 05, 06, and 07.

Normative semantic rules

37. External-domain exchange MUST be modeled through an External Trust Domain, an applicable Treaty, and the Constitutional Firewall.
38. Every active Treaty MUST be exact, scoped, time-bounded, revocable, auditable, and explicitly Owner-authorized through the Owner Authorization Ceremony bound to that exact Treaty.
39. A Treaty MUST NOT grant Authority prohibited by Book I or bypass constitutional enforcement.
40. Data use MUST state classification, authorized purpose, minimization, access, retention, disclosure, and disposal rules.
41. Security Controls and Authority Controls MUST be distinguished even when implemented by the same mechanism.

Canonical term set

HAL-TERM-0116 — External Trust Domain

An external environment, organization, system, or authority regime whose controls and assumptions are not natively governed by HAL.

Required distinction: External status does not imply hostility or trustworthiness; exchange requires explicit governance.

Type: External governance domain | Category: Trust | Status: Approved | Aliases: ETD

Source basis: Direct source normalization

Book I: Book I Decision 49 | Book II: Book II Chapters 20 and 21 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0116` when it uses **External Trust Domain** with this exact governed meaning: An external environment, organization, system, or authority regime whose controls and assumptions are not natively governed by HAL.

*Counterexample: A dependent artifact uses **External Trust Domain** in a way that violates its required distinction: External status does not imply hostility or trustworthiness; exchange requires explicit governance.*

Relationships: HAL-REL-0041, HAL-REL-0050 | Lifecycle: None registered

Constraint: External status does not imply hostility or trustworthiness; exchange requires explicit governance.

HAL-TERM-0117 — Treaty

An exact, scoped, time-bounded, revocable, auditable, and explicitly Owner-authorized agreement defining allowed cross-domain identities, data, capabilities, authority, obligations, verification, monitoring, failure behavior, and termination.

Required distinction: Activation requires the Owner Authorization Ceremony bound to the exact Treaty. Conversation, delegated ordinary authority, trust, usefulness, prior access, or the Treaty itself cannot grant authority prohibited by Book I or bypass the Constitutional Firewall.

Type: Governed trust agreement | Category: Trust | Status: Approved | Aliases: External Trust Treaty

Source basis: Direct source normalization

Book I: Book I Decision 49 | Book II: Book II Chapter 21 | Book III: Book III Chapters 5 and 7

Example: A dependent artifact cites `HAL-TERM-0117` when it uses **Treaty** with this exact governed meaning: An exact, scoped, time-bounded, revocable, auditable, and explicitly Owner-authorized agreement defining allowed cross-domain identities, data, capabilities, authority, obligations, verification, monitoring, failure behavior, and termination.

*Counterexample: A dependent artifact uses **Treaty** in a way that violates its required distinction: Activation requires the Owner Authorization Ceremony bound to the exact Treaty. Conversation, delegated ordinary authority, trust, usefulness, prior access, or the Treaty itself cannot grant authority prohibited by Book I or bypass the Constitutional Firewall.*

Relationships: HAL-REL-0041, HAL-REL-0042 | Lifecycle: HAL-TRANS-0018, HAL-TRANS-0019, HAL-TRANS-0020

Constraint: Activation requires the Owner Authorization Ceremony bound to the exact Treaty. Conversation, delegated ordinary authority, trust, usefulness, prior access, or the Treaty itself cannot grant authority prohibited by Book I or bypass the Constitutional Firewall.

HAL-TERM-0118 — Constitutional Firewall

The Book II boundary that mediates and enforces constitutionally governed exchange between HAL and External Trust Domains.

Required distinction: It is not merely a network firewall and must not be bypassed by direct integration.

Type: Architectural enforcement boundary | Category: Trust | Status: Approved | Aliases: CF

Source basis: Direct source normalization

Book I: Book I Decision 49 | Book II: Book II Chapter 20 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0118` when it uses **Constitutional Firewall** with this exact governed meaning: The Book II boundary that mediates and enforces constitutionally governed exchange between HAL and External Trust Domains.

*Counterexample: A dependent artifact uses **Constitutional Firewall** in a way that violates its required distinction: It is not merely a network firewall and must not be bypassed by direct integration.*

Relationships: HAL-REL-0042 | Lifecycle: None registered

Constraint: It is not merely a network firewall and must not be bypassed by direct integration.

HAL-TERM-0119 — Trust Boundary

A boundary across which identity, authority, integrity, confidentiality, provenance, or governance assumptions change.

Required distinction: Network adjacency alone does not define or erase a Trust Boundary.

Type: Security boundary | Category: Security | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 26, 27, 32, 37, 39, 48, and 49 | Book II: Book II Chapters 18, 19, 20, 21, 25, and 26 | Book III: Book III Chapters 2, 4, 5, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0119` when it uses **Trust Boundary** with this exact governed meaning: A boundary across which identity, authority, integrity, confidentiality, provenance, or governance assumptions change.

*Counterexample: A dependent artifact uses **Trust Boundary** in a way that violates its required distinction: Network adjacency alone does not define or erase a Trust Boundary.*

Relationships: None registered | Lifecycle: None registered

Constraint: Network adjacency alone does not define or erase a Trust Boundary.

HAL-TERM-0120 — Data Classification

A governed label determining handling, access, transmission, retention, evidence, and disposal requirements for information.

Required distinction: Classification is not a purpose or permission to use the data.

Type: Governance label | Category: Privacy | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 32 | Book II: Book II Chapter 19 | Book III: Book III Chapters 4 and 5

Example: A dependent artifact cites `HAL-TERM-0120` when it uses **Data Classification** with this exact governed meaning: A governed label determining handling, access, transmission, retention, evidence, and disposal requirements for information.

*Counterexample: A dependent artifact uses **Data Classification** in a way that violates its required distinction: Classification is not a purpose or permission to use the data.*

Relationships: HAL-REL-0043 | Lifecycle: None registered

Constraint: Classification is not a purpose or permission to use the data.

HAL-TERM-0121 — Personal Data

Information relating to an identified or reasonably identifiable human under the governing privacy context.

Required distinction: Pseudonymization may reduce exposure but does not necessarily remove personal-data status.

Type: Information class | Category: Privacy | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 32 | Book II: Book II Chapter 19 | Book III: Book III Chapters 4 and 5

Example: A dependent artifact cites `HAL-TERM-0121` when it uses **Personal Data** with this exact governed meaning: Information relating to an identified or reasonably identifiable human under the governing privacy context.

*Counterexample: A dependent artifact uses **Personal Data** in a way that violates its required distinction: Pseudonymization may reduce exposure but does not necessarily remove personal-data status.*

Relationships: None registered | Lifecycle: None registered

Constraint: Pseudonymization may reduce exposure but does not necessarily remove personal-data status.

HAL-TERM-0122 — Sensitive Data

Information whose unauthorized use or disclosure could materially harm a person, HAL, the Owner, security, sovereignty, trust, or protected operations.

Required distinction: Sensitivity is context-dependent and may include non-personal operational data.

Type: Information class | Category: Privacy | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 32 | Book II: Book II Chapter 19 | Book III: Book III Chapters 4 and 5

Example: A dependent artifact cites `HAL-TERM-0122` when it uses **Sensitive Data** with this exact governed meaning: Information whose unauthorized use or disclosure could materially harm a person, HAL, the Owner, security, sovereignty, trust, or protected operations.

*Counterexample: A dependent artifact uses **Sensitive Data** in a way that violates its required distinction: Sensitivity is context-dependent and may include non-personal operational data.*

Relationships: None registered | Lifecycle: None registered

Constraint: Sensitivity is context-dependent and may include non-personal operational data.

HAL-TERM-0123 — Purpose Limitation

The rule that governed data may be collected, used, shared, and retained only for declared, authorized, compatible purposes.

Required distinction: Availability or technical usefulness does not establish purpose.

Type: Use constraint | Category: Privacy | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 32 | Book II: Book II Chapter 19 | Book III: Book III Chapters 4 and 5

Example: A dependent artifact cites `HAL-TERM-0123` when it uses **Purpose Limitation** with this exact governed meaning: The rule that governed data may be collected, used, shared, and retained only for declared, authorized, compatible purposes.

*Counterexample: A dependent artifact uses **Purpose Limitation** in a way that violates its required distinction: Availability or technical usefulness does not establish purpose.*

Relationships: None registered | Lifecycle: None registered

Constraint: Availability or technical usefulness does not establish purpose.

HAL-TERM-0124 — Data Minimization

The rule that only the least data reasonably necessary for an authorized purpose may be collected, processed, retained, and disclosed.

Required distinction: Minimization applies to fields, precision, population, duration, access, and derived inferences.

Type: Collection and use constraint | Category: Privacy | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 32 | Book II: Book II Chapter 19 | Book III: Book III Chapters 4 and 5

Example: A dependent artifact cites `HAL-TERM-0124` when it uses **Data Minimization** with this exact governed meaning: The rule that only the least data reasonably necessary for an authorized purpose may be collected, processed, retained, and disclosed.

*Counterexample: A dependent artifact uses **Data Minimization** in a way that violates its required distinction: Minimization applies to fields, precision, population, duration, access, and derived inferences.*

Relationships: None registered | Lifecycle: None registered

Constraint: Minimization applies to fields, precision, population, duration, access, and derived inferences.

HAL-TERM-0125 — Retention Class

A governed category defining retention period, review, legal or constitutional hold behavior, archival, and disposal requirements.

Required distinction: A Retention Class does not itself authorize collection or access.

Type: Lifecycle label | Category: Privacy | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decision 32 | Book II: Book II Chapter 19 | Book III: Book III Chapters 4 and 5

Example: A dependent artifact cites `HAL-TERM-0125` when it uses **Retention Class** with this exact governed meaning: A governed category defining retention period, review, legal or constitutional hold behavior, archival, and disposal requirements.

*Counterexample: A dependent artifact uses **Retention Class** in a way that violates its required distinction: A Retention Class does not itself authorize collection or access.*

Relationships: HAL-REL-0043 | Lifecycle: None registered

Constraint: A Retention Class does not itself authorize collection or access.

HAL-TERM-0126 — Secret

Confidential material whose disclosure could enable unauthorized access, impersonation, decryption, signing, or protected operation.

Required distinction: An Identifier or public key is not a Secret merely because it is security-related.

Type: Sensitive authentication material | Category: Security | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 27, 37, and 39 | Book II: Book II Chapters 05, 19, and 26 | Book III: Book III Chapters 2 and 5

Example: A dependent artifact cites `HAL-TERM-0126` when it uses **Secret** with this exact governed meaning: Confidential material whose disclosure could enable unauthorized access, impersonation, decryption, signing, or protected operation.

*Counterexample: A dependent artifact uses **Secret** in a way that violates its required distinction: An Identifier or public key is not a Secret merely because it is security-related.*

Relationships: None registered | Lifecycle: None registered

Constraint: An Identifier or public key is not a Secret merely because it is security-related.

HAL-TERM-0127 — Cryptographic Key

A governed value used by an approved cryptographic operation and managed through generation, storage, access, rotation, revocation, destruction, and provenance controls.

Required distinction: A key's possession does not itself establish business Authority.

Type: Cryptographic material | Category: Security | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 27, 37, and 39 | Book II: Book II Chapters 05, 19, and 26 | Book III: Book III Chapters 2 and 5

Example: A dependent artifact cites `HAL-TERM-0127` when it uses **Cryptographic Key** with this exact governed meaning: A governed value used by an approved cryptographic operation and managed through generation, storage, access, rotation, revocation, destruction, and provenance controls.

*Counterexample: A dependent artifact uses **Cryptographic Key** in a way that violates its required distinction: A key's possession does not itself establish business Authority.*

Relationships: None registered | Lifecycle: None registered

Constraint: A key's possession does not itself establish business Authority.

HAL-TERM-0128 — Security Control

A safeguard intended to protect confidentiality, integrity, availability, identity assurance, or system resilience.

Required distinction: A Security Control protecting HAL is distinct from an Authority Control preventing HAL from exceeding its mandate.

Type: Protective control | Category: Security | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 27, 37, and 39 | Book II: Book II Chapters 05, 19, and 26 | Book III: Book III Chapters 2 and 5

Example: A dependent artifact cites `HAL-TERM-0128` when it uses **Security Control** with this exact governed meaning: A safeguard intended to protect confidentiality, integrity, availability, identity assurance, or system resilience.

*Counterexample: A dependent artifact uses **Security Control** in a way that violates its required distinction: A Security Control protecting HAL is distinct from an Authority Control preventing HAL from exceeding its mandate.*

Relationships: None registered | Lifecycle: None registered

Constraint: A Security Control protecting HAL is distinct from an Authority Control preventing HAL from exceeding its mandate.

HAL-TERM-0129 — Authority Control

A safeguard that prevents a Principal, component, or HAL from deciding or acting beyond governed Authority.

Required distinction: It is not interchangeable with a Security Control, though one mechanism may support both.

Type: Mandate-limiting control | Category: Authority | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 27, 37, and 39 | Book II: Book II Chapters 05, 19, and 26 | Book III: Book III Chapters 2 and 5

Example: A dependent artifact cites `HAL-TERM-0129` when it uses **Authority Control** with this exact governed meaning: A safeguard that prevents a Principal, component, or HAL from deciding or acting beyond governed Authority.

*Counterexample: A dependent artifact uses **Authority Control** in a way that violates its required distinction: It is not interchangeable with a Security Control, though one mechanism may support both.*

Relationships: None registered | Lifecycle: None registered

Constraint: It is not interchangeable with a Security Control, though one mechanism may support both.

HAL-TERM-0163 — Trust Domain

A bounded identity, policy, evidence, security, privacy, and accountability context within which trust assumptions and controls are evaluated.

Required distinction: Trust Domain is the generic concept. An External Trust Domain is a Trust Domain outside HAL's native governance boundary and requires governed cross-domain exchange.

Type: Governance context | Category: Trust | Status: Approved | Aliases: None

Source basis: Derived semantic synthesis required by the Book X scope

Book I: Book I Decisions 49 and 52 | Book II: Book II Chapters 18, 21, and 24 | Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0163` when it uses **Trust Domain** with this exact governed meaning: A bounded identity, policy, evidence, security, privacy, and accountability context within which trust assumptions and controls are evaluated.

*Counterexample: A dependent artifact uses **Trust Domain** in a way that violates its required distinction: Trust Domain is the generic concept. An External Trust Domain is a Trust Domain outside HAL's native governance boundary and requires governed cross-domain exchange.*

Relationships: HAL-REL-0050 | Lifecycle: None registered

Constraint: Trust Domain is the generic concept. An External Trust Domain is a Trust Domain outside HAL's native governance boundary and requires governed cross-domain exchange.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining External Trust Domain locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 11 — Runtime, Resources, Operations, Failure, Recovery, and Change

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines runtime state, health, readiness, supervision, resources, degradation, incidents, recovery objectives, releases, migrations, and exceptions. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I continuity, reversibility, restraint, accountability, and constitutional shutdown.

Book II: Book II Chapters 02, 22, 27, 28, 29, 33, 34, and 35.

Book III: Book III Chapters 01, 02, 04, 06, 07, 08, and 09.

Normative semantic rules

42. Runtime health MUST distinguish Liveness, Readiness, Health, Desired State, and Observed State.
43. Degraded, Safe, Restricted, Quarantined, and Recovering modes MUST be explicit, observable, and governed by transition criteria.
44. Recovery MUST restore identity, authority, state, evidence, and trust as applicable; restart alone MUST NOT be called Recovery.
45. Exceptions MUST be time-bounded and MUST NOT waive Constitutional Invariants; Architecture Deviations MUST use architecture governance.

Canonical term set

HAL-TERM-0130 — Runtime

The governed combination of processes, state, resources, policies, dependencies, and environments in which HAL behavior executes.

Required distinction: A Runtime is not HAL's constitutional identity.

Type: Execution environment | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0130` when it uses **Runtime** with this exact governed meaning: The governed combination of processes, state, resources, policies, dependencies, and environments in which HAL behavior executes.

*Counterexample: A dependent artifact uses **Runtime** in a way that violates its required distinction: A Runtime is not HAL's constitutional identity.*

Relationships: None registered | Lifecycle: None registered

Constraint: A Runtime is not HAL's constitutional identity.

HAL-TERM-0131 — Node

A governed compute or device participant capable of hosting workload, state, sensing, or action under the runtime architecture.

Required distinction: A Node does not independently become HAL or own authoritative state without explicit designation.

Type: Execution entity | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0131` when it uses **Node** with this exact governed meaning: A governed compute or device participant capable of hosting workload, state, sensing, or action under the runtime architecture.

*Counterexample: A dependent artifact uses **Node** in a way that violates its required distinction: A Node does not independently become HAL or own authoritative state without explicit designation.*

Relationships: None registered | Lifecycle: None registered

Constraint: A Node does not independently become HAL or own authoritative state without explicit designation.

HAL-TERM-0132 — Service

A deployable runtime boundary providing one or more governed responsibilities or interfaces.

Required distinction: A Service is not automatically an architectural component, capability, or authority domain.

Type: Component deployment role | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0132` when it uses **Service** with this exact governed meaning: A deployable runtime boundary providing one or more governed responsibilities or interfaces.

*Counterexample: A dependent artifact uses **Service** in a way that violates its required distinction: A Service is not automatically an architectural component, capability, or authority domain.*

Relationships: HAL-REL-0044 | Lifecycle: HAL-TRANS-0021, HAL-TRANS-0022, HAL-TRANS-0023, HAL-TRANS-0024

Constraint: A Service is not automatically an architectural component, capability, or authority domain.

HAL-TERM-0133 — Supervisor

A component that monitors and governs workload lifecycle, desired state, health, restart, containment, and escalation within its authority.

Required distinction: A Supervisor must not conceal repeated failure through unlimited restart loops.

Type: Runtime control role | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0133` when it uses **Supervisor** with this exact governed meaning: A component that monitors and governs workload lifecycle, desired state, health, restart, containment, and escalation within its authority.

*Counterexample: A dependent artifact uses **Supervisor** in a way that violates its required distinction: A Supervisor must not conceal repeated failure through unlimited restart loops.*

Relationships: HAL-REL-0044 | Lifecycle: None registered

Constraint: A Supervisor must not conceal repeated failure through unlimited restart loops.

HAL-TERM-0134 — Desired State

The governed runtime condition an authorized controller intends the system to maintain.

Required distinction: Desired State is not proof of Observed State or successful convergence.

Type: Control target | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0134` when it uses **Desired State** with this exact governed meaning: The governed runtime condition an authorized controller intends the system to maintain.

*Counterexample: A dependent artifact uses **Desired State** in a way that violates its required distinction: Desired State is not proof of Observed State or successful convergence.*

Relationships: HAL-REL-0046 | Lifecycle: None registered

Constraint: Desired State is not proof of Observed State or successful convergence.

HAL-TERM-0135 — Observed State

The evidenced runtime condition measured at a defined time and scope.

Required distinction: Observation may be stale, partial, or uncertain and must not be treated as Desired State.

Type: Runtime observation | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0135` when it uses **Observed State** with this exact governed meaning: The evidenced runtime condition measured at a defined time and scope.

*Counterexample: A dependent artifact uses **Observed State** in a way that violates its required distinction: Observation may be stale, partial, or uncertain and must not be treated as Desired State.*

Relationships: None registered | Lifecycle: None registered

Constraint: Observation may be stale, partial, or uncertain and must not be treated as Desired State.

HAL-TERM-0136 — Health

A multidimensional assessment of a component's ability to perform its declared responsibilities within current constraints.

Required distinction: Process liveness alone is not Health.

Type: Operational assessment | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0136` when it uses **Health** with this exact governed meaning: A multidimensional assessment of a component's ability to perform its declared responsibilities within current constraints.

*Counterexample: A dependent artifact uses **Health** in a way that violates its required distinction: Process liveness alone is not Health.*

Relationships: None registered | Lifecycle: None registered

Constraint: Process liveness alone is not Health.

HAL-TERM-0137 — Liveness

Evidence that a workload is running or able to make progress according to a narrow declared probe.

Required distinction: Liveness does not establish readiness, correctness, authorization, or safety.

Type: Runtime signal | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0137` when it uses **Liveness** with this exact governed meaning: Evidence that a workload is running or able to make progress according to a narrow declared probe.

*Counterexample: A dependent artifact uses **Liveness** in a way that violates its required distinction: Liveness does not establish readiness, correctness, authorization, or safety.*

Relationships: None registered | Lifecycle: None registered

Constraint: Liveness does not establish readiness, correctness, authorization, or safety.

HAL-TERM-0138 — Readiness

Evidence that a workload may receive its declared class of work under current dependencies, configuration, and safety conditions.

Required distinction: Readiness is scoped and must not be inferred from Liveness.

Type: Admission signal | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0138` when it uses **Readiness** with this exact governed meaning: Evidence that a workload may receive its declared class of work under current dependencies, configuration, and safety conditions.

*Counterexample: A dependent artifact uses **Readiness** in a way that violates its required distinction: Readiness is scoped and must not be inferred from Liveness.*

Relationships: None registered | Lifecycle: None registered

Constraint: Readiness is scoped and must not be inferred from Liveness.

HAL-TERM-0139 — Resource

A bounded consumable or allocatable asset such as compute, memory, storage, bandwidth, attention, time, energy, or device capacity.

Required distinction: Resource availability does not imply authority to allocate or consume it.

Type: Governed capacity | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0139` when it uses **Resource** with this exact governed meaning: A bounded consumable or allocatable asset such as compute, memory, storage, bandwidth, attention, time, energy, or device capacity.

*Counterexample: A dependent artifact uses **Resource** in a way that violates its required distinction: Resource availability does not imply authority to allocate or consume it.*

Relationships: HAL-REL-0045 | Lifecycle: None registered

Constraint: Resource availability does not imply authority to allocate or consume it.

HAL-TERM-0140 — Reservation

A time-bounded governed allocation claim against a Resource for a declared purpose and owner.

Required distinction: A Reservation is not proof the resource was consumed or the work completed.

Type: Resource claim | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0140` when it uses **Reservation** with this exact governed meaning: A time-bounded governed allocation claim against a Resource for a declared purpose and owner.

*Counterexample: A dependent artifact uses **Reservation** in a way that violates its required distinction: A Reservation is not proof the resource was consumed or the work completed.*

Relationships: HAL-REL-0045 | Lifecycle: None registered

Constraint: A Reservation is not proof the resource was consumed or the work completed.

HAL-TERM-0141 — Degraded Mode

A declared operating state in which selected capabilities or service levels are reduced while higher-priority constitutional, authority, safety, privacy, and evidence obligations remain protected.

Required distinction: Degradation must not silently weaken non-degradable controls.

Type: Operating state | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0141` when it uses **Degraded Mode** with this exact governed meaning: A declared operating state in which selected capabilities or service levels are reduced while higher-priority constitutional, authority, safety, privacy, and evidence obligations remain protected.

*Counterexample: A dependent artifact uses **Degraded Mode** in a way that violates its required distinction: Degradation must not silently weaken non-degradable controls.*

Relationships: None registered | Lifecycle: None registered

Constraint: Degradation must not silently weaken non-degradable controls.

HAL-TERM-0142 — Safe Mode

A constrained operating state that prioritizes containment, Owner communication, essential evidence, and prevention of unauthorized or unsafe effects.

Required distinction: Safe Mode is not a generic low-performance mode.

Type: Protective operating state | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0142` when it uses **Safe Mode** with this exact governed meaning: A constrained operating state that prioritizes containment, Owner communication, essential evidence, and prevention of unauthorized or unsafe effects.

*Counterexample: A dependent artifact uses **Safe Mode** in a way that violates its required distinction: Safe Mode is not a generic low-performance mode.*

Relationships: None registered | Lifecycle: None registered

Constraint: Safe Mode is not a generic low-performance mode.

HAL-TERM-0143 — Restricted Mode

An operating state in which selected capabilities, integrations, or authority paths are disabled or narrowed because required trust, evidence, policy, or assurance is unavailable.

Required distinction: It must be explicit, observable, and reversible through governed recovery.

Type: Authority-limited state | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51 | Book II: Book II Chapters 02, 27, 28, 33, and 34 | Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0143` when it uses **Restricted Mode** with this exact governed meaning: An operating state in which selected capabilities, integrations, or authority paths are disabled or narrowed because required trust, evidence, policy, or assurance is unavailable.

*Counterexample: A dependent artifact uses **Restricted Mode** in a way that violates its required distinction: It must be explicit, observable, and reversible through governed recovery.*

Relationships: None registered | Lifecycle: None registered

Constraint: It must be explicit, observable, and reversible through governed recovery.

HAL-TERM-0144 — Incident

An event or condition requiring coordinated response because it threatens constitutional conformance, authority, security, privacy, trust, availability, integrity, or material outcomes.

Required distinction: An anomaly is not necessarily an Incident until classification criteria are met.

Type: Governed adverse event | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50 | Book II: Book II Chapters 27 and 28 | Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0144` when it uses **Incident** with this exact governed meaning: An event or condition requiring coordinated response because it threatens constitutional conformance, authority, security, privacy, trust, availability, integrity, or material outcomes.

*Counterexample: A dependent artifact uses **Incident** in a way that violates its required distinction: An anomaly is not necessarily an Incident until classification criteria are met.*

Relationships: None registered | Lifecycle: None registered

Constraint: An anomaly is not necessarily an Incident until classification criteria are met.

HAL-TERM-0145 — Quarantine

A governed isolation state preventing a component, artifact, identity, message, or data set from participating beyond an explicitly limited inspection boundary.

Required distinction: Quarantine is not deletion and must preserve required evidence.

Type: Containment state | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50 | Book II: Book II Chapters 27 and 28 | Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0145` when it uses **Quarantine** with this exact governed meaning: A governed isolation state preventing a component, artifact, identity, message, or data set from participating beyond an explicitly limited inspection boundary.

*Counterexample: A dependent artifact uses **Quarantine** in a way that violates its required distinction: Quarantine is not deletion and must preserve required evidence.*

Relationships: None registered | Lifecycle: None registered

Constraint: Quarantine is not deletion and must preserve required evidence.

HAL-TERM-0146 — Recovery

The governed process of restoring acceptable identity, authority, state, service, evidence, and trust after failure or compromise.

Required distinction: Restart alone is not Recovery.

Type: Restoration process | Category: Operations | Status: Approved | Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50 | Book II: Book II Chapters 27 and 28 | Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0146` when it uses **Recovery** with this exact governed meaning: The governed process of restoring acceptable identity, authority, state, service, evidence, and trust after failure or compromise.

*Counterexample: A dependent artifact uses **Recovery** in a way that violates its required distinction: Restart alone is not Recovery.*

Relationships: HAL-REL-0046 | Lifecycle: None registered

Constraint: Restart alone is not Recovery.

HAL-TERM-0147 — Recovery Point Objective

The maximum tolerable loss or unavailability of recoverable state measured from the disruption point for a declared domain.

Required distinction: RPO is a target, not proof that recovery met it.

Type: Recovery target | Category: Operations | Status: Approved | Aliases: RPO

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50 | Book II: Book II Chapters 27 and 28 | Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0147` when it uses **Recovery Point Objective** with this exact governed meaning: The maximum tolerable loss or unavailability of recoverable state measured from the disruption point for a declared domain.

*Counterexample: A dependent artifact uses **Recovery Point Objective** in a way that violates its required distinction: RPO is a target, not proof that recovery met it.*

Relationships: None registered | Lifecycle: None registered

Constraint: RPO is a target, not proof that recovery met it.

HAL-TERM-0148 — Recovery Time Objective

The target duration for restoring a declared service or capability to its required state after disruption.

Required distinction: RTO does not authorize unsafe shortcuts during recovery.

Type: Recovery target | Category: Operations | Status: Approved | Aliases: RTO

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50 | Book II: Book II Chapters 27 and 28 | Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0148` when it uses **Recovery Time Objective** with this exact governed meaning: The target duration for restoring a declared service or capability to its required state after disruption.

*Counterexample: A dependent artifact uses **Recovery Time Objective** in a way that violates its required distinction: RTO does not authorize unsafe shortcuts during recovery.*

Relationships: None registered | Lifecycle: None registered

Constraint: RTO does not authorize unsafe shortcuts during recovery.

HAL-TERM-0149 — Release

A versioned, traceable, qualified set of software, configuration, schemas, models, and deployment artifacts approved for a declared environment and scope.

Required distinction: A successful build is not a Release.

Type: Governed artifact set | Category: Engineering | Status: Approved | Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0149` when it uses **Release** with this exact governed meaning: A versioned, traceable, qualified set of software, configuration, schemas, models, and deployment artifacts approved for a declared environment and scope.

*Counterexample: A dependent artifact uses **Release** in a way that violates its required distinction: A successful build is not a Release.*

Relationships: HAL-REL-0054 | Lifecycle: HAL-TRANS-0025, HAL-TRANS-0026

Constraint: A successful build is not a Release.

HAL-TERM-0150 — Migration

A governed transition of data, state, schema, interface, configuration, or runtime behavior from one compatible condition to another.

Required distinction: A Migration must distinguish reversible steps from irreversible effects and compensation.

Type: State or contract change | Category: Engineering | Status: Approved | Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0150` when it uses **Migration** with this exact governed meaning: A governed transition of data, state, schema, interface, configuration, or runtime behavior from one compatible condition to another.

*Counterexample: A dependent artifact uses **Migration** in a way that violates its required distinction: A Migration must distinguish reversible steps from irreversible effects and compensation.*

Relationships: None registered | Lifecycle: None registered

Constraint: A Migration must distinguish reversible steps from irreversible effects and compensation.

HAL-TERM-0151 — Architecture Decision Record

A durable record of a consequential architecture decision, context, alternatives, rationale, consequences, source traceability, and review status.

Required distinction: An ADR cannot authorize deviation from Book II without the approved architecture-governance process.

Type: Decision record | Category: Engineering | Status: Approved | Aliases: ADR

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0151` when it uses **Architecture Decision Record** with this exact governed meaning: A durable record of a consequential architecture decision, context, alternatives, rationale, consequences, source traceability, and review status.

*Counterexample: A dependent artifact uses **Architecture Decision Record** in a way that violates its required distinction: An ADR cannot authorize deviation from Book II without the approved architecture-governance process.*

Relationships: HAL-REL-0047 | Lifecycle: None registered

Constraint: An ADR cannot authorize deviation from Book II without the approved architecture-governance process.

HAL-TERM-0152 — Architecture Deviation

A documented departure from an applicable Book II requirement processed through architecture governance with scope, risk, evidence, approval, and expiry or remediation.

Required distinction: It is not an ordinary Book III control exception and cannot amend Book II silently.

Type: Governed exception class | Category: Engineering | Status: Approved | Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0152` when it uses **Architecture Deviation** with this exact governed meaning: A documented departure from an applicable Book II requirement processed through architecture governance with scope, risk, evidence, approval, and expiry or remediation.

*Counterexample: A dependent artifact uses **Architecture Deviation** in a way that violates its required distinction: It is not an ordinary Book III control exception and cannot amend Book II silently.*

Relationships: None registered | Lifecycle: None registered

Constraint: It is not an ordinary Book III control exception and cannot amend Book II silently.

HAL-TERM-0153 — Control

A stable, attributable requirement with applicability, responsibility, enforcement, evidence, severity, exception authority, verification, and source traceability.

Required distinction: Advice is not a Control unless it is made objectively reviewable.

Type: Enforceable rule record | Category: Engineering | Status: Approved | Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0153` when it uses **Control** with this exact governed meaning: A stable, attributable requirement with applicability, responsibility, enforcement, evidence, severity, exception authority, verification, and source traceability.

*Counterexample: A dependent artifact uses **Control** in a way that violates its required distinction: Advice is not a Control unless it is made objectively reviewable.*

Relationships: HAL-REL-0048 | Lifecycle: None registered

Constraint: Advice is not a Control unless it is made objectively reviewable.

HAL-TERM-0154 — Exception

A documented, scoped, risk-assessed, compensating, approved, expiring departure from a waivable lower-order Control.

Required distinction: A constitutional invariant cannot be waived; an Architecture Deviation uses its own governance path.

Type: Time-bounded control relief | Category: Engineering | Status: Approved | Aliases: waiver

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0154` when it uses **Exception** with this exact governed meaning: A documented, scoped, risk-assessed, compensating, approved, expiring departure from a waivable lower-order Control.

*Counterexample: A dependent artifact uses **Exception** in a way that violates its required distinction: A constitutional invariant cannot be waived; an Architecture Deviation uses its own governance path.*

Relationships: HAL-REL-0048 | Lifecycle: HAL-TRANS-0027, HAL-TRANS-0028

Constraint: A constitutional invariant cannot be waived; an Architecture Deviation uses its own governance path.

HAL-TERM-0155 — Definition of Ready

The minimum evidenced conditions required before consequential implementation work may begin or enter its next controlled stage.

Required distinction: Readiness does not imply approval to release or cross the Reality Boundary.

Type: Entry criteria | Category: Engineering | Status: Approved | Aliases: DoR

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0155` when it uses **Definition of Ready** with this exact governed meaning: The minimum evidenced conditions required before consequential implementation work may begin or enter its next controlled stage.

*Counterexample: A dependent artifact uses **Definition of Ready** in a way that violates its required distinction: Readiness does not imply approval to release or cross the Reality Boundary.*

Relationships: None registered | Lifecycle: None registered

Constraint: Readiness does not imply approval to release or cross the Reality Boundary.

HAL-TERM-0156 — Definition of Done

The minimum evidenced conditions required before work may be treated as complete within a declared scope.

Required distinction: Done does not erase post-release monitoring, retention, or recovery obligations.

Type: Completion criteria | Category: Engineering | Status: Approved | Aliases: DoD

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0156` when it uses **Definition of Done** with this exact governed meaning: The minimum evidenced conditions required before work may be treated as complete within a declared scope.

*Counterexample: A dependent artifact uses **Definition of Done** in a way that violates its required distinction: Done does not erase post-release monitoring, retention, or recovery obligations.*

Relationships: None registered | Lifecycle: None registered

Constraint: Done does not erase post-release monitoring, retention, or recovery obligations.

HAL-TERM-0166 — Release Authority

The Book III-designated role authorized to certify a qualified Release for a declared deployment scope after all required verification and reviews have passed.

Required distinction: Release Authority cannot waive Constitutional Invariants, replace architecture, security, or privacy review, or authorize deployment beyond the certified scope.

Type: Governed certification role | Category: Engineering | Status: Approved | Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58 | Book II: Book II Chapters 29 and 35 | Book III: Book III Chapter 7

Example: The Release Authority certifies Release `R-42` only after the scoped qualification evidence and required architecture, security, and privacy reviews pass.

Counterexample: A successful CI build deploys itself because pipeline success is treated as Release Authority approval.

Relationships: HAL-REL-0054 | Lifecycle: None registered

Constraint: Release Authority cannot waive Constitutional Invariants, replace architecture, security, or privacy review, or authorize deployment beyond the certified scope.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Runtime locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Chapter 12 — Naming, Acronyms, Ambiguity, Deprecation, and Cross-Book Use

Final v1.0 | Authority: Books I → II → III → X

Purpose and scope

Defines canonical labels, aliases, acronyms, qualified terms, forbidden ambiguities, semantic versioning, deprecation, and adoption rules. It governs shared meaning and does not independently create component behavior or interface syntax.

Source authority

Book I: Book I constitutional authority and evolution.

Book II: Book II Chapters 29, 30, and 35.

Book III: Book III Chapters 01, 03, 04, 08, and 09.

Normative semantic rules

46. Canonical Labels MUST be used in normative canon text; aliases MAY be used only when meaning remains unambiguous.
47. Acronyms MUST be registered and expanded on first use unless the artifact's audience and scope make the expansion unambiguous.
48. A term MUST NOT be silently repurposed; incompatible meaning requires a new term or a major Semantic Version with migration.
49. Forbidden and deprecated usages MUST identify the safer replacement and adoption path.

Canonical term set

HAL-TERM-0157 — Canonical Label

The single approved label used as the primary reference for one Canonical Term.

Required distinction: Capitalization is part of controlled usage when needed to distinguish the term from ordinary language.

Type: Naming element | Category: Naming | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29, 30, and 35 | Book III: Book III Chapters 1, 3, 4, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0157` when it uses **Canonical Label** with this exact governed meaning: The single approved label used as the primary reference for one Canonical Term.

*Counterexample: A dependent artifact uses **Canonical Label** in a way that violates its required distinction: Capitalization is part of controlled usage when needed to distinguish the term from ordinary language.*

Relationships: None registered | Lifecycle: None registered

Constraint: Capitalization is part of controlled usage when needed to distinguish the term from ordinary language.

HAL-TERM-0158 — Acronym

An approved shortened form mapped to exactly one canonical expansion within its declared scope.

Required distinction: An Acronym must be expanded on first use unless the audience and artifact make the meaning unambiguous.

Type: Abbreviated label | Category: Naming | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29, 30, and 35 | Book III: Book III Chapters 1, 3, 4, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0158` when it uses **Acronym** with this exact governed meaning: An approved shortened form mapped to exactly one canonical expansion within its declared scope.

*Counterexample: A dependent artifact uses **Acronym** in a way that violates its required distinction: An Acronym must be expanded on first use unless the audience and artifact make the meaning unambiguous.*

Relationships: None registered | Lifecycle: None registered

Constraint: An Acronym must be expanded on first use unless the audience and artifact make the meaning unambiguous.

HAL-TERM-0159 — Semantic Version

A version assigned to the Book X corpus to communicate the compatibility impact of semantic changes.

Required distinction: It does not replace the versioning of Books I-III, components, or interfaces.

Type: Compatibility marker | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29, 30, and 35 | Book III: Book III Chapters 1, 3, 4, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0159` when it uses **Semantic Version** with this exact governed meaning: A version assigned to the Book X corpus to communicate the compatibility impact of semantic changes.

*Counterexample: A dependent artifact uses **Semantic Version** in a way that violates its required distinction: It does not replace the versioning of Books I-III, components, or interfaces.*

Relationships: None registered | Lifecycle: None registered

Constraint: It does not replace the versioning of Books I-III, components, or interfaces.

HAL-TERM-0160 — Term Status

The controlled state of a Term Record: Proposed, Candidate, Approved, Deprecated, Retired, or Rejected.

Required distinction: Status must not be inferred from document age or usage frequency.

Type: Lifecycle label | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29, 30, and 35 | Book III: Book III Chapters 1, 3, 4, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0160` when it uses **Term Status** with this exact governed meaning: The controlled state of a Term Record: Proposed, Candidate, Approved, Deprecated, Retired, or Rejected.

*Counterexample: A dependent artifact uses **Term Status** in a way that violates its required distinction: Status must not be inferred from document age or usage frequency.*

Relationships: None registered | Lifecycle: None registered

Constraint: Status must not be inferred from document age or usage frequency.

HAL-TERM-0161 — Cross-Book Term Index

The mapping from each Canonical Term to its authoritative source, Book X record, and known use across the HAL canon.

Required distinction: It is an index, not a substitute for reading the governing source.

Type: Traceability index | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29, 30, and 35 | Book III: Book III Chapters 1, 3, 4, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0161` when it uses **Cross-Book Term Index** with this exact governed meaning: The mapping from each Canonical Term to its authoritative source, Book X record, and known use across the HAL canon.

*Counterexample: A dependent artifact uses **Cross-Book Term Index** in a way that violates its required distinction: It is an index, not a substitute for reading the governing source.*

Relationships: HAL-REL-0049 | Lifecycle: None registered

Constraint: It is an index, not a substitute for reading the governing source.

HAL-TERM-0162 — Semantic Compatibility

The condition in which a terminology or information-model change preserves the valid interpretation and obligations of dependent artifacts within declared scope.

Required distinction: Textual similarity alone does not establish Semantic Compatibility.

Type: Compatibility relation | Category: Governance | Status: Approved | Aliases: None

Source basis: Book X semantic-governance choice constrained by higher-order sources

Book I: Book I Constitutional Governance and Decision 58 | Book II: Book II Chapters 29, 30, and 35 | Book III: Book III Chapters 1, 3, 4, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0162` when it uses **Semantic Compatibility** with this exact governed meaning: The condition in which a terminology or information-model change preserves the valid interpretation and obligations of dependent artifacts within declared scope.

*Counterexample: A dependent artifact uses **Semantic Compatibility** in a way that violates its required distinction: Textual similarity alone does not establish Semantic Compatibility.*

Relationships: None registered | Lifecycle: None registered

Constraint: Textual similarity alone does not establish Semantic Compatibility.

Relationship and lifecycle rules

Relationships involving these terms MUST use the typed relationship records in the Entity Relationship Register. Lifecycle-bearing concepts MUST use the transition vocabulary in the Lifecycle State Register; a status label alone MUST NOT imply that an otherwise prohibited transition is valid.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit and does not imply an undocumented relationship or transition.

Anti-patterns

Semantic drift: redefining Canonical Label locally without a governed Semantic Change.

Authority laundering: using a definition, alias, schema field, or component name to imply authority not granted by Books I–III.

Representation collapse: treating an entity, its identifier, its record, and its current state as interchangeable.

Verification

Verify stable-ID and Canonical-Label uniqueness, source traceability, source-basis classification, non-circular definitions, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-book compatibility.

Change and deprecation

A proposed change MUST include source authority, compatibility classification, dependent-artifact impact, migration guidance, reviewer, effective version, and—when deprecating—a replacement plus sunset condition. Book X maintainers may resolve routine lexical and modeling matters but MUST escalate only if the change would interpret constitutional philosophy or alter an Owner-reserved matter.

Review findings

The corrected chapter passed constitutional fidelity, architecture fidelity, engineering fidelity, semantic consistency, clarity, usability, machine readability, cross-format parity, and Owner-threshold review.

Owner Review items and completion

None. Complete and approved for Book X v1.0.

Appendix A — Complete glossary index

Term ID	Canonical label	Type	Chapter
HAL-TERM-0001	Book X	Canonical reference	1
HAL-TERM-0002	Canonical Term	Semantic record	1
HAL-TERM-0003	Term Record	Metadata record	1
HAL-TERM-0004	Semantic Authority	Precedence property	1
HAL-TERM-0005	Semantic Change	Governed change	1
HAL-TERM-0006	Qualified Term	Disambiguated label	1
HAL-TERM-0007	Allowed Alias	Reference label	1
HAL-TERM-0008	Deprecated Term	Lifecycle status	1
HAL-TERM-0009	Forbidden Term	Prohibited label or usage	1
HAL-TERM-0010	Normative Source	Source role	1
HAL-TERM-0011	Entity	Semantic type	2
HAL-TERM-0012	Value Object	Semantic type	2
HAL-TERM-0013	Record	Semantic type	2
HAL-TERM-0014	Authoritative Record	State role	2
HAL-TERM-0015	Relationship	Semantic type	2
HAL-TERM-0016	State	Semantic type	2
HAL-TERM-0017	Lifecycle	Semantic model	2
HAL-TERM-0018	Invariant	Constraint	2
HAL-TERM-0019	Claim	Evidence-bearing assertion	2
HAL-TERM-0020	Constraint	Rule element	2
HAL-TERM-0021	Owner	Constitutional role	3
HAL-TERM-0022	HAL	Constitutional identity	3
HAL-TERM-0023	Constitution	Supreme governing instrument	3
HAL-TERM-0024	Constitutional Invariant	Constitutional constraint	3
HAL-TERM-0025	Constitutional Kernel	Architectural component class	3
HAL-TERM-0026	Constitutional Mirror	Self-description mechanism	3
HAL-TERM-0027	Continuity	Constitutional property	3
HAL-TERM-0028	Presence	Contextual manifestation	3
HAL-TERM-0029	Embodiment	Contextual binding	3
HAL-TERM-0030	Sovereignty	Constitutional property	3
HAL-TERM-0031	Identity	Governed entity identity	4
HAL-TERM-0032	Principal	Governed actor role	4
HAL-TERM-0033	Identity Record	Authoritative record	4

HAL-TERM-0034	Identifier	Reference value	4
HAL-TERM-0035	Identity Attribute	Governed descriptive value	4
HAL-TERM-0036	Credential	Authentication instrument	4
HAL-TERM-0037	Authentication	Assurance process and result	4
HAL-TERM-0038	Authentication Evidence	Evidence role	4
HAL-TERM-0039	Trust	Evidence-based assessment	4
HAL-TERM-0040	Permission	Decision result	4
HAL-TERM-0041	Authority	Governed decision and action scope	4
HAL-TERM-0042	Delegation	Governed authority grant	4
HAL-TERM-0043	Policy	Decision rule set	4
HAL-TERM-0044	Policy Decision Record	Decision record	4
HAL-TERM-0045	Protected Action	Risk classification	4
HAL-TERM-0046	Intent	Purpose object	5
HAL-TERM-0047	Vision	Long-horizon intent	5
HAL-TERM-0048	Goal	Outcome target	5
HAL-TERM-0049	Objective	Measurable target	5
HAL-TERM-0050	Project	Coordinated work container	5
HAL-TERM-0051	Task	Work unit	5
HAL-TERM-0052	Strategy	Approach selection	5
HAL-TERM-0053	Plan	Coordinated intended work	5
HAL-TERM-0054	Plan Graph	Planning representation	5
HAL-TERM-0055	Execution Graph	Runtime representation	5
HAL-TERM-0056	Attention Object	Prioritization record	5
HAL-TERM-0057	Decision Object	Decision record	5
HAL-TERM-0058	Judgment	Reasoned evaluation	5
HAL-TERM-0059	Uncertainty	Epistemic condition	5
HAL-TERM-0060	Outcome Object	Outcome record	5
HAL-TERM-0061	Success	Evaluated condition	5
HAL-TERM-0062	Capability	Implementation-independent contract	6
HAL-TERM-0063	Capability Contract	Contract record	6
HAL-TERM-0064	Provider	Implementation role	6
HAL-TERM-0065	Adapter	Boundary component role	6
HAL-TERM-0066	Capability Registry	Authoritative registry	6
HAL-TERM-0067	Action	State-changing attempt	6

HAL-TERM-0068	Attempt	Execution instance	6
HAL-TERM-0069	Transaction	Governed action lifecycle	6
HAL-TERM-0070	Commit Barrier	Irreversibility gate	6
HAL-TERM-0071	Rollback	Reversal operation	6
HAL-TERM-0072	Compensation	Remedial operation	6
HAL-TERM-0073	Reality Boundary	Governed environment boundary	6
HAL-TERM-0074	Simulation	Non-reality environment	6
HAL-TERM-0075	Digital Twin	Modeled counterpart	6
HAL-TERM-0076	Shadow Execution	Non-committing execution mode	6
HAL-TERM-0077	Canary	Limited reality stage	6
HAL-TERM-0078	Evidence Candidate	Pre-admission record	7
HAL-TERM-0079	Evidence Object	Immutable provenance-bearing record	7
HAL-TERM-0080	Audit Record	Protected accountability record	7
HAL-TERM-0081	Provenance	Origin history	7
HAL-TERM-0082	Chain of Custody	Integrity history	7
HAL-TERM-0083	Evidence Graph	Evidence relationship model	7
HAL-TERM-0084	Verification	Evidence-producing evaluation	7
HAL-TERM-0085	Verification Plan	Planning record	7
HAL-TERM-0086	Assurance Case	Structured argument	7
HAL-TERM-0087	Certification	Governed assurance decision	7
HAL-TERM-0088	Conformance	Evaluated relation	7
HAL-TERM-0089	Confidence	Calibrated assessment	7
HAL-TERM-0090	Experience	Retained occurrence record	8
HAL-TERM-0091	Experience Ledger	Authoritative ledger	8
HAL-TERM-0092	Memory	Retrievable retained representation	8
HAL-TERM-0093	Memory Graph	Associative representation	8
HAL-TERM-0094	Knowledge	Contextualized warranted representation	8
HAL-TERM-0095	Knowledge Graph	Knowledge relationship model	8
HAL-TERM-0096	Pattern	Learned regularity	8
HAL-TERM-0097	Learning	Governed update process	8
HAL-TERM-0098	Learning Ledger	Change ledger	8
HAL-TERM-0099	Wisdom	Evidence-bounded judgment resource	8
HAL-TERM-0100	Self Model	Governed self-representation	8
HAL-TERM-0101	Command	Intent message	9
HAL-TERM-0102	Query	Read request	9

HAL-TERM-0103	Event	Completed-fact record	9
HAL-TERM-0104	Event Journal	Durable event store	9
HAL-TERM-0105	Message Envelope	Transport record	9
HAL-TERM-0106	Idempotency Key	Deduplication value	9
HAL-TERM-0107	Correlation Identifier	Trace reference	9
HAL-TERM-0108	Causation Identifier	Causal reference	9
HAL-TERM-0109	Authoritative State	Owned state	9
HAL-TERM-0110	Projection	Derived state	9
HAL-TERM-0111	Cache	Disposable derived state	9
HAL-TERM-0112	Replica	Replicated state	9
HAL-TERM-0113	Transactional Outbox	Publication pattern	9
HAL-TERM-0114	Logical Time	Ordering construct	9
HAL-TERM-0115	Wall-Clock Time	Temporal observation	9
HAL-TERM-0116	External Trust Domain	External governance domain	10
HAL-TERM-0117	Treaty	Governed trust agreement	10
HAL-TERM-0118	Constitutional Firewall	Architectural enforcement boundary	10
HAL-TERM-0119	Trust Boundary	Security boundary	10
HAL-TERM-0120	Data Classification	Governance label	10
HAL-TERM-0121	Personal Data	Information class	10
HAL-TERM-0122	Sensitive Data	Information class	10
HAL-TERM-0123	Purpose Limitation	Use constraint	10
HAL-TERM-0124	Data Minimization	Collection and use constraint	10
HAL-TERM-0125	Retention Class	Lifecycle label	10
HAL-TERM-0126	Secret	Sensitive authentication material	10
HAL-TERM-0127	Cryptographic Key	Cryptographic material	10
HAL-TERM-0128	Security Control	Protective control	10
HAL-TERM-0129	Authority Control	Mandate-limiting control	10
HAL-TERM-0130	Runtime	Execution environment	11
HAL-TERM-0131	Node	Execution entity	11
HAL-TERM-0132	Service	Component deployment role	11
HAL-TERM-0133	Supervisor	Runtime control role	11
HAL-TERM-0134	Desired State	Control target	11
HAL-TERM-0135	Observed State	Runtime observation	11
HAL-TERM-0136	Health	Operational assessment	11
HAL-TERM-0137	Liveness	Runtime signal	11

HAL-TERM-0138	Readiness	Admission signal	11
HAL-TERM-0139	Resource	Governed capacity	11
HAL-TERM-0140	Reservation	Resource claim	11
HAL-TERM-0141	Degraded Mode	Operating state	11
HAL-TERM-0142	Safe Mode	Protective operating state	11
HAL-TERM-0143	Restricted Mode	Authority-limited state	11
HAL-TERM-0144	Incident	Governed adverse event	11
HAL-TERM-0145	Quarantine	Containment state	11
HAL-TERM-0146	Recovery	Restoration process	11
HAL-TERM-0147	Recovery Point Objective	Recovery target	11
HAL-TERM-0148	Recovery Time Objective	Recovery target	11
HAL-TERM-0149	Release	Governed artifact set	11
HAL-TERM-0150	Migration	State or contract change	11
HAL-TERM-0151	Architecture Decision Record	Decision record	11
HAL-TERM-0152	Architecture Deviation	Governed exception class	11
HAL-TERM-0153	Control	Enforceable rule record	11
HAL-TERM-0154	Exception	Time-bounded control relief	11
HAL-TERM-0155	Definition of Ready	Entry criteria	11
HAL-TERM-0156	Definition of Done	Completion criteria	11
HAL-TERM-0157	Canonical Label	Naming element	12
HAL-TERM-0158	Acronym	Abbreviated label	12
HAL-TERM-0159	Semantic Version	Compatibility marker	12
HAL-TERM-0160	Term Status	Lifecycle label	12
HAL-TERM-0161	Cross-Book Term Index	Traceability index	12
HAL-TERM-0162	Semantic Compatibility	Compatibility relation	12
HAL-TERM-0163	Trust Domain	Governance context	10
HAL-TERM-0164	Owner Authorization Ceremony	Protected authorization mechanism	4
HAL-TERM-0165	Evidence Service	Architectural component class	7
HAL-TERM-0166	Release Authority	Governed certification role	11

Appendix B — Relationship catalog

HAL-REL-0001

Owner — owns constitutionally → HAL. Cardinality: 1 to 1. Book I-reserved ownership; infrastructure possession is insufficient.

HAL-REL-0002

Constitution — governs → HAL. Cardinality: 1 to 1. Supreme authority.

HAL-REL-0003

Constitutional Kernel — enforces → Constitutional Invariant. Cardinality: 1 to many. Only at Book II-designated enforcement points.

HAL-REL-0004

Constitutional Mirror — describes → HAL. Cardinality: 1 to 1. Evidence-linked and non-self-authorizing.

HAL-REL-0005

HAL — manifests through → Presence. Cardinality: 1 to many. All Presences share one HAL identity.

HAL-REL-0006

Presence — binds to → Embodiment. Cardinality: many to 0..many. Within explicit context and lifecycle.

HAL-REL-0007

Identity Record — represents → Identity. Cardinality: 1 to 1. Record and entity remain distinct.

HAL-REL-0008

Identifier — references → Identity. Cardinality: many to 1. Within a declared namespace.

HAL-REL-0009

Identity Attribute — describes → Identity. Cardinality: many to 1. Does not establish authority.

HAL-REL-0010

Authentication Evidence — supports → Authentication. Cardinality: many to many. Evidence role, not authorization.

HAL-REL-0011

Delegation — grants bounded → Authority. Cardinality: many to 1. Cannot exceed delegator authority.

HAL-REL-0012

Policy — evaluates for → Permission. Cardinality: many to many. Result is contextual and scoped.

HAL-REL-0013

Authority — constrains → Permission. Cardinality: many to many. Authority is not the decision result.

HAL-REL-0014

Trust — informs → Policy Decision Record. Cardinality: many to many. Never sole authority.

HAL-REL-0015

Intent — decomposes into → Goal. Cardinality: 1 to many. Traceability is retained.

HAL-REL-0016

Goal — decomposes into → Objective. Cardinality: 1 to many. Criteria remain explicit.

HAL-REL-0017

Objective — is pursued by → Plan. Cardinality: many to many. Plans do not confer authority.

HAL-REL-0018

Plan — contains → Task. Cardinality: 1 to many. Execution may diverge with evidence.

HAL-REL-0019

Decision Object — records → Judgment. Cardinality: many to 1. Includes alternatives and uncertainty.

HAL-REL-0020

Outcome Object — evaluates → Goal. Cardinality: many to many. Against evidence and side effects.

HAL-REL-0021

Capability Contract — defines → Capability. Cardinality: many to 1. Versioned semantic contract.

HAL-REL-0022

Provider — fulfills → Capability. Cardinality: many to many. Selection remains governed.

HAL-REL-0023

Adapter — connects → Provider. Cardinality: many to 1. Preserves canonical capability semantics.

HAL-REL-0024

Transaction — coordinates → Action. Cardinality: 1 to many. Includes commit and recovery state.

HAL-REL-0025

Canary — is governed stage within → Reality Boundary. Cardinality: many to 1. A Canary is a limited real-operation stage; it is not a kind of boundary.

HAL-REL-0026

Evidence Candidate — may be admitted as → Evidence Object. Cardinality: many to 0..1. Only through the authoritative evidence process.

HAL-REL-0027

Evidence Object — supports or opposes → Claim. Cardinality: many to many. Relation and weight are explicit.

HAL-REL-0028

Evidence Graph — contains → Evidence Object. Cardinality: 1 to many. Objects remain immutable.

HAL-REL-0029

Verification — evaluates → Claim. Cardinality: many to many. Against explicit criteria.

HAL-REL-0030

Assurance Case — organizes → Claim. Cardinality: 1 to many. Includes reasoning and defeaters.

HAL-REL-0031

Certification — depends on → Assurance Case. Cardinality: many to 1..many. Scoped and time-bounded.

HAL-REL-0032

Experience Ledger — contains → Experience. Cardinality: 1 to many. Append-oriented and governed.

HAL-REL-0033

Memory Graph — associates → Memory. Cardinality: 1 to many. Association is not causation.

HAL-REL-0034

Knowledge Graph — represents → Knowledge. Cardinality: 1 to many. Preserves provenance and validity.

HAL-REL-0035

Pattern — is derived from → Experience. Cardinality: many to many. Reproducibly supported.

HAL-REL-0036

Wisdom — informs → Judgment. Cardinality: many to many. Does not grant authority.

HAL-REL-0037

Command — may cause → Event. Cardinality: many to 0..many. Only after authoritative handling.

HAL-REL-0038

Projection — is derived from → Event Journal. Cardinality: many to 1..many. Not authoritative unless designated.

HAL-REL-0039

Transactional Outbox — publishes → Event. Cardinality: 1 to many. After local authoritative commit.

HAL-REL-0040

Message Envelope — carries → Command. Cardinality: many to 0..1. Also may carry Query or Event.

HAL-REL-0041

Treaty — governs exchange with → External Trust Domain. Cardinality: many to 1. Revocable and constitutionally bounded.

HAL-REL-0042

Constitutional Firewall — enforces → Treaty. Cardinality: 1 to many. At external exchange paths.

HAL-REL-0043

Data Classification — constrains → Retention Class. Cardinality: many to many. Alongside purpose and authority.

HAL-REL-0044

Supervisor — controls lifecycle of → Service. Cardinality: many to many. Within declared authority.

HAL-REL-0045

Reservation — allocates → Resource. Cardinality: many to 1. Time-bounded and purpose-bound.

HAL-REL-0046

Recovery — restores toward → Desired State. Cardinality: many to 1. Validated against Observed State.

HAL-REL-0047

Architecture Decision Record — documents → Semantic Change. Cardinality: many to 0..many. When architectural consequence exists.

HAL-REL-0048

Exception — applies to → Control. Cardinality: many to 1. Time-bounded; never constitutional.

HAL-REL-0049

Cross-Book Term Index — indexes → Canonical Term. Cardinality: 1 to many. Does not replace governing source.

HAL-REL-0050

External Trust Domain — specializes → Trust Domain. Cardinality: many to 1. Externality changes governance assumptions and requires controlled exchange.

HAL-REL-0051

Owner — performs → Owner Authorization Ceremony. Cardinality: 1 to many. Each ceremony is bound to one exact immutable decision identifier, scope, and validity period.

HAL-REL-0052

Owner Authorization Ceremony — authorizes exact → Protected Action. Cardinality: many to 1. Authorization is non-transferable and cannot be reused for a different action, Treaty, capability class, or mutation.

HAL-REL-0053

Evidence Service — admits and governs → Evidence Object. Cardinality: 1 to many. Only the authoritative admission process may create governed Evidence Objects or change their verification state through new linked evidence.

HAL-REL-0054

Release Authority — certifies → Release. Cardinality: many to many. Certification is evidence-based, attributable, scoped, and cannot exceed the qualified release or deployment scope.

Appendix C — Lifecycle transition catalog

HAL-TRANS-0001 — Term Record: Proposed → Candidate

Entry condition: Semantic steward accepts a complete proposal. Required evidence: Proposal and source mapping.

HAL-TRANS-0002 — Term Record: Candidate → Approved

Entry condition: Cross-book review passes and authority is confirmed. Required evidence: Review record and decision.

HAL-TRANS-0003 — Term Record: Approved → Deprecated

Entry condition: Replacement and migration plan are approved. Required evidence: Deprecation notice.

HAL-TRANS-0004 — Term Record: Deprecated → Retired

Entry condition: Sunset conditions are met and dependents are migrated. Required evidence: Retirement verification.

HAL-TRANS-0005 — Delegation: Draft → Active

Entry condition: Authorized delegator signs within scope. Required evidence: Delegation record.

HAL-TRANS-0006 — Delegation: Active → Expired

Entry condition: Expiration time is reached. Required evidence: Expiry event.

HAL-TRANS-0007 — Delegation: Active → Revoked

Entry condition: Authorized revoker acts. Required evidence: Revocation evidence.

HAL-TRANS-0008 — Transaction: Proposed → Authorized

Entry condition: Authority and policy checks pass. Required evidence: Policy Decision Record.

HAL-TRANS-0009 — Transaction: Authorized → Prepared

Entry condition: Preconditions and resources are secured. Required evidence: Preparation evidence.

HAL-TRANS-0010 — Transaction: Prepared → Committed

Entry condition: Commit Barrier conditions pass. Required evidence: Commit record.

HAL-TRANS-0011 — Transaction: Committed → Completed

Entry condition: Effects and outcomes are observed. Required evidence: Outcome and Evidence Objects.

HAL-TRANS-0012 — Transaction: Prepared → Rolled Back

Entry condition: Reversible state is restored. Required evidence: Rollback evidence.

HAL-TRANS-0013 — Transaction: Committed → Compensating

Entry condition: Irreversible effects require remediation. Required evidence: Compensation decision.

HAL-TRANS-0014 — Evidence Candidate: Collected → Admitted

Entry condition: Evidence Service validates provenance and custody. Required evidence: Admission record.

HAL-TRANS-0015 — Certification: Proposed → Active

Entry condition: Authorized certifier approves scoped assurance case. Required evidence: Certification record.

HAL-TRANS-0016 — Certification: Active → Suspended

Entry condition: Material evidence defect or risk invalidates reliance. Required evidence: Suspension record.

HAL-TRANS-0017 — Certification: Active → Expired

Entry condition: Validity period ends. Required evidence: Expiry record.

HAL-TRANS-0018 — Treaty: Draft → Active

Entry condition: The Owner Authorization Ceremony approves the exact, time-bounded Treaty. Required evidence: Owner authorization bound to the exact Treaty plus Constitutional Firewall activation evidence.

HAL-TRANS-0019 — Treaty: Active → Suspended

Entry condition: Boundary conditions or trust fail. Required evidence: Suspension and containment record.

HAL-TRANS-0020 — Treaty: Active → Revoked

Entry condition: Authorized party terminates the Treaty. Required evidence: Revocation evidence.

HAL-TRANS-0021 — Service: Starting → Ready

Entry condition: Readiness criteria pass. Required evidence: Readiness observation.

HAL-TRANS-0022 — Service: Ready → Degraded

Entry condition: Declared service conditions fall below threshold. Required evidence: Health and incident evidence.

HAL-TRANS-0023 — Service: Degraded → Quarantined

Entry condition: Containment criteria require isolation. Required evidence: Quarantine record.

HAL-TRANS-0024 — Service: Quarantined → Recovering

Entry condition: Recovery plan is authorized. Required evidence: Recovery record.

HAL-TRANS-0025 — Release: Candidate → Qualified

Entry condition: Required verification and reviews pass. Required evidence: Release-readiness record.

HAL-TRANS-0026 — Release: Qualified → Released

Entry condition: Release Authority approves deployment scope. Required evidence: Release certification.

HAL-TRANS-0027 — Exception: Proposed → Active

Entry condition: Authorized approver accepts bounded residual risk. Required evidence: Exception record.

HAL-TRANS-0028 — Exception: Active → Expired

Entry condition: Expiration date is reached. Required evidence: Fail-closed signal or escalation.

Appendix D — Acronyms and ambiguity

Approved acronyms

ADR — Architecture Decision Record. Engineering and architecture decisions

API — Application Programming Interface. Interface reference

CF — Constitutional Firewall. Trust architecture

DoD — Definition of Done. Engineering lifecycle

DoR — Definition of Ready. Engineering lifecycle

ETD — External Trust Domain. Trust architecture

HAL — HAL. Constitutional identity; not expanded into an invented phrase

ID — Identifier. Use only where the namespace is clear

PII — Personally Identifiable Information. Prefer Personal Data in canonical prose unless a legal regime requires PII

RPO — Recovery Point Objective. Recovery

RTO — Recovery Time Objective. Recovery

SBOM — Software Bill of Materials. Supply-chain evidence

SDLC — Secure Development Lifecycle. Engineering

SLO — Service Level Objective. Operations

TOC — Table of Contents. Publication

Forbidden or qualification-required usages

user

Use: Use Principal, Owner, human, operator, or another qualified role. Reason: “User” collapses distinct identity and authority roles.

agent

Use: Use HAL, Principal, service, model, provider, or external agent. Reason: “Agent” obscures identity, accountability, and authority.

authorization

Use: Use Authority for governed scope; Permission for the decision result; Policy evaluation for the process. Reason: The word often collapses three distinct concepts.

proof

Use: Use Evidence Object, Verification result, or formal proof as applicable. Reason: Evidence supports claims; empirical evidence is not necessarily mathematical proof.

truth

Use: Use authoritative state, verified claim, observation, or confidence-qualified conclusion. Reason: Unqualified truth hides source, time, scope, and uncertainty.

memory

Use: Use Experience, Experience Ledger, Memory, Knowledge, or cache as applicable. Reason: The generic word hides governance, durability, and epistemic status.

production

Use: Qualify the exact environment and Reality Boundary stage. Reason: A name does not establish real authority or effect boundaries.

rollback

Use: Use Rollback only for truthful reversal; use Compensation for remedial new action. Reason: External effects may not be erasable.

exactly once

Use: State the bounded delivery, deduplication, and effect guarantee. Reason: Distributed and external effects rarely support an unqualified guarantee.

real time

Use: Declare latency, freshness, clock, and ordering bounds. Reason: The phrase is not objectively testable without thresholds.

secure

Use: Name the control objective, threat, enforcement, and evidence. Reason: A broad adjective is not a security claim.

safe

Use: Name the hazard, invariant, containment, verification, and residual risk. Reason: A broad adjective is not a safety claim.

trusted

Use: Name the Trust dimension, scope, evidence, confidence, and expiry. Reason: Trust is multidimensional and does not imply authority.

owner

Use: Capitalize Owner only for the constitutional role; qualify other ownership such as code owner or data custodian. Reason: Lowercase operational ownership must not be confused with Book I authority.

HAL instance

Use: Use Runtime, Node, Presence, service instance, or model instance. Reason: HAL has one constitutional identity.

evidence

Use: Use Evidence Object when authoritative admission is meant; otherwise qualify Evidence Candidate or source material. Reason: Not every record or observation is authoritative Evidence.

Founder

Use: Use Owner in new canon text; Founder is permitted only as a historical source alias for that same role. Reason: Book I states Founder and Owner are the same constitutional role; Founder must not be interpreted as a second role.

Appendix E — Cross-book adoption rules

1. Books I–III remain controlling and are never rewritten merely to match Book X.

2. Books IV–IX MUST use Book X stable IDs and Canonical Labels when they mean a Book X concept.

3. Component-specific terms belong in Book IV but SHOULD reuse or specialize Book X concepts without redefining them.

4. Machine-facing contract names belong in Book IX and MUST map to Book X terms where the semantics are shared.

5. Operations, security, governance, and verification manuals MAY introduce domain procedures but MUST NOT repurpose Book X labels.

6. A dependent artifact encountering an ambiguity MUST qualify the term, cite the Term ID, and submit a Semantic Change proposal when the canonical corpus is insufficient.

Certification status

Final v1.0. 166 terms, 54 relationships, 28 lifecycle transitions. No open Owner Review item.